

Unraveling the Success of Poland and the Challenges Impeding Ukraine's Growth

Bachelor Thesis

Supervisor:

Prof. Ing. Lubor Lacina, Ph.D.

Anna Liamina

Brno 2025

Acknowledgments

I would like to thank my supervisor, Prof. Ing. Lubor Lacina, Ph.D. for the support, feedback, and guidance provided throughout the process of writing this thesis. His involvement helped me stay focused and gradually move forward at each stage of the work.

I also wish to express my appreciation to the faculty members of Mendel University for the opportunity to study in this academic environment and for the knowledge and experience I have gained during my studies. This period has contributed significantly to my academic and personal development.

Finally, I would like to acknowledge the support of my parents and close ones. Although we were apart for most of this time, their support and encouragement remained important and meaningful throughout the entire studying journey.

Statutory declaration

I hereby declare that, this thesis entitled: **Unraveling the Success of Poland and the Challenges Impeding Ukraine's Growth** was written and completed by me. I also declare that all the sources and information used to complete the thesis are included in the list of references. I agree that the thesis could be made public in accordance with Article 47b of Act No. 111/1998 Coll., Higher Education Institutions and on Amendments and Supplements to Some Other Acts (the Higher Education Act), and in accordance with the current Directive on publishing of the final thesis.

I am aware that my thesis is written in accordance to Act. 121/2000 Coll., on Copyright and therefore Mendel University in Brno has the right to conclude licence agreements on the utilization of the thesis as a school work in accordance with Article 60(1) of the Copyright Act.

Before concluding a licence agreement on utilization of the work by another person, I will request a written statement from the university that the licence agreement is not in contradiction to legitimate interests of the university, and I will also pay a prospective fee to cover the cost incurred in creating the work to the full amount of such costs.

In Brno on 3th May, 2025

Abstract

Since the fall of the Soviet Union, both Ukraine and Poland have faced significant challenges and transformations. Despite starting from similar economic conditions in the early 1990s, Poland has achieved substantial economic growth, particularly through European Union (EU) integration, while Ukraine has struggled with internal and external challenges that have slowed its progress. This thesis aims to identify the key factors that enabled Poland to surpass Ukraine in economic development, despite their comparable starting points. Through a comparative analysis of economic indicators such as GDP growth, inflation, and unemployment rates, the study focuses on the period from approximately 1989 to 2023. The empirical research will analyze time series data for both countries to highlight differences in economic performance. The findings indicate that Poland's timely EU integration, successful market reforms, and strong institutional frameworks facilitated its transition to a market-driven economy. Conversely, Ukraine's economic development has been hindered by political instability, incomplete reforms, and corruption.

Keywords: Economic transition, comparative analysis, market reforms, success factors, EU integration, Poland, Ukraine

Abstrakt

Od rozpadu Sovětského svazu čelily Ukrajina i Polsko významným výzvám a proměnám. Přestože obě země začínaly na počátku 90. let z podobných ekonomických podmínek, Polsko dosáhlo výrazného hospodářského růstu, zejména díky integraci do Evropské unie (EU), zatímco Ukrajina čelila vnitřním i vnějším problémům, které zpomalily její pokrok. Tato diplomová práce si klade za cíl identifikovat klíčové faktory, které umožnily Polsku překonat Ukrajinu v hospodářském rozvoji, přestože obě země měly srovnatelné výchozí podmínky. Pomocí srovnávací analýzy ekonomických ukazatelů, jako jsou růst HDP, inflace a míra nezaměstnanosti, se studie zaměřuje na období přibližně od roku 1989 do roku 2023. Empirický výzkum analyzuje časové řady dat obou zemí, aby zdůraznil rozdíly v ekonomické výkonnosti. Zjištění ukazují, že včasná integrace Polska do EU, úspěšné tržní reformy a silné institucionální rámce umožnily jeho přechod na tržně orientovanou ekonomiku. Naopak hospodářský rozvoj Ukrajiny byl brzděn politickou nestabilitou, neúplnými reformami a korupcí.

Klíčová slova: ekonomická transformace, srovnávací analýza, tržní reformy, faktory úspěchu, integrace do EU, Polsko, Ukrajina

Contents

Introduction and objectives	9
1 Literature review	11
1.1 Review of Academic Perspectives on Transition Economies.....	11
1.2 Economic Transition in Poland: Path to Success.....	14
1.3 Economic development challenges in Ukraine.....	18
1.4 Comparative Economic Studies Between Poland and Ukraine	23
1.5 Partial Conclusions Based on Literature Review	26
2 Methods and data.....	28
3 Empirical analysis.....	30
3.1 Overview of Basic Economic Indicators	30
3.2 Institutional quality and governance	37
3.3 Trade and investments patterns	50
3.4 Economic Growth and Development Barriers: A Comparative Visualization	59
4 Recommendations	67
5 Discussion and conclusion.....	69
6 References.....	71
List of figures	74
List of tables.....	76
Attachments.....	77
A Tables with data used in the empirical part.....	78
B Econometric models used in empirical part.....	84

Introduction and objectives

Although Poland and Ukraine share a common historical experience under the Soviet influence, they followed completely different economic trajectories. Both countries transitioned from centrally planned economies to market-driven systems, yet their paths and outcomes have been remarkably different. This divergence began in the late 1980s, when Soviet General Secretary Mikhail Gorbachev introduced the economic restructuring policy known as "perestroika." While both Poland and Ukraine began shifting towards market reforms, their levels of dependence on the Soviet system shaped each country's transition processes in distinct ways (Boyarchuk, 2019).

Today, Poland is recognized as one of the most successful post-socialist economies, that achieved economic stability and growth, whereas Ukraine has faced persistent challenges.

Poland's economic reforms, particularly those introduced by Leszek Balcerowicz, played a critical role in its transition. The implementation of "shock therapy" in the early 1990s led to short-term economic hardship but laid the foundation for long-term growth. This rapid liberalization included price deregulation, reduction of state intervention, and mass privatization, allowing Poland to establish a market economy relatively quickly (Svejnar, 2002).

Ukraine, in contrast, pursued a slower and less consistent reform path. Following the collapse of the Soviet Union, economic restructuring was delayed by political instability, corruption, and weak institutional development. Many analysts argue that a more decisive approach to reform—such as the one taken by Poland—could have accelerated Ukraine's economic progress. However, political factors, including the election of Leonid Kravchuk over reform-oriented Vyacheslav Chornovil in 1991, significantly influenced the trajectory of economic policy (Breslauer, 2021). Additionally, Ukraine's economic development was further disrupted by the 2014 Russian invasion of Donbas and Luhansk, the annexation of Crimea, and the full-scale war in 2022.

Poland's accession to the EU in 2004 was another decisive moment, as it provided access to financial resources, institutional guidance, and an expanded market. According to the World Bank, Poland's GDP grew at an average annual rate of 4.2% between 1990 and 2019, driven by industrial expansion, increased exports, and foreign direct investment (FDI). In contrast, Ukraine's GDP growth was more volatile due to political crises, military conflicts, and persistent corruption. Between 1991 and 2019, Ukraine's GDP growth rate averaged only 1.2% annually (World Bank, 2023).

This thesis draws on data from sources such as the World Bank, Eurostat, national statistics offices, and other relevant institutions. Through a comparison of essential economic indicators for Ukraine and Poland, the study seeks to explore how different institutional paths and policy approaches have shaped the countries' current economic conditions.

The main objective of this research is to run a comparative analysis of the economic trajectories of Poland and Ukraine over the period from 1988 to 2023, depending on available statistical data. The study focuses on identifying the

institutional and policy factors that supported Poland's dynamic growth and contrasts them with the challenges that have slowed Ukraine's development down.

To comprehensively address this topic, the thesis will explore the following research questions:

1. What were the key challenges faced by Poland and Ukraine during their economic transitions, and how did these challenges differ?
2. What factors led to Poland's economic success, and how can they help shape policy recommendations to support Ukraine's economic development?

The study is structured into three main parts, each designed to offer a deeper understanding of the issue. The first part provides a historical and literature overview of economic transition processes. The second part presents a comparative analysis of economic data from both countries, highlighting the impact of policy decisions. The final section explores policy recommendations that could help Ukraine strengthen its economic stability.

The topic of this thesis was chosen due to the author's interest in the economic development of their home country. By analyzing the factors behind Poland's success and Ukraine's ongoing struggles, this research aims to contribute to a better understanding of the necessary steps for fostering sustainable economic growth in Ukraine.

1 Literature review

1.1 Review of Academic Perspectives on Transition Economies

The change from a centrally planned economy to a market-based system became a key challenge for post-communist countries in the late 20th century. Generally, a transition economy is an economic system undergoing transformation from central planning to market mechanisms, a process accompanied by extensive institutional, political, and social changes (Douarin & Mickiewicz, 2017). This shift includes price liberalization, privatization, macroeconomic stabilization, and the restructuring of legal and market institutions necessary for the functioning of a competitive economy based on private ownership (Svejnar, 2002).

When it comes to specific time periods authors tend to attribute the period of transition economies to the late 1980s and early 2000s. Svejnar (2002) examines the first 12 years following the collapse of communist regimes, starting in 1989. Fidrmuc (2001) analyzes the transition process during 1990–1993, a period marked by key macroeconomic changes. Douarin & Mickiewicz (2017) and Gevorkyan (2018) emphasize that institutional reforms extended well into the 2000s, with some nations experiencing prolonged transitions, particularly in terms of legal system development and the establishment of market institutions. Therefore, depending on the specific fields of study the period could be extended even farther away.

Price liberalization was considered a necessary step to eliminate imbalances left after the planned economy. The removal of price controls led to a sharp increase in inflation but facilitated the establishment of market-based pricing mechanisms (Svejnar, 2002). Macroeconomic stabilization involved measures to control inflation, maintain exchange rate stability, and regulate budget deficits (Sergi, 2020). Despite initial economic difficulties, the implementation of strict fiscal and monetary policies contributed to stabilization and prepared the foundation for further growth.

Privatization of state-owned assets became a central element of economic transformation. The primary approaches included mass voucher privatization, direct sales to investors, and the transfer of enterprises to management and employees. The effectiveness of these methods mostly depended on the development level of institutional environments and legal regulations (Fidrmuc, 2001). In some countries, weak control over privatization processes led to increased corruption and the dawn of oligarchic structures, limiting economic growth potential. According to Berenson (2018), successful economic transition depended not only on market reforms but also on the profound restructuring of state bureaucracies.

Institutional reforms were essential for establishing a stable market environment. Mickiewicz (2017) notes that the development of legal institutions, protection of property rights, and effective corporate governance were key factors in successful economic transformation. Informal institutions, such as social norms and public expectations, also played a crucial role. In countries with high levels of trust in government institutions, reforms were implemented more consistently, whereas regions with low trust in authorities faced significant resistance to market adaptation. Moreover, the public's attitude toward the former regime and its level of awareness regarding the benefits of a market economy played a significant role in shaping the transition process.

Social perceptions, whether favoring the old system or embracing economic liberalization, influenced both the pace of reforms and their acceptance by the population.

Political stability also impacted economic transition. Democratization contributed to the creation of favorable conditions for investment and private sector development. However, in the early stages of reform, it could introduce additional sources of instability (Fidrmuc, 2001). Long-term studies indicate that the consistency of reforms and political predictability positively impacted economic growth.

The European Union played a crucial role in the economic transition process by encouraging institutional reforms in Central and Eastern European countries. Sergi (2020) examines the EU's impact on economic transformations, introducing the concept of the "Brussels Consensus" as an alternative to the Washington Consensus. The Washington Consensus prioritized swift market liberalization and strict fiscal policies, while the Brussels Consensus promoted a more comprehensive, institution-focused strategy that combined economic and governance reforms within the broader context of European integration. The EU provided clear institutional development guidelines, helping countries align their economic policies with European standards. This approach allowed integrating nations to achieve greater macroeconomic stability and accelerate economic growth. In addition, international financial organizations, such as the International Monetary Fund and the World Bank, supported transition economies under the condition of implementing strict fiscal and structural reforms.

Overall, an analysis of economic transition suggests that sustainable development is impossible without profound institutional changes. While macroeconomic stabilization is necessary in the early stages, long-term success requires an effective legal environment, well-developed market institutions, and integration into international economic systems. The experience of transition economies confirms that a comprehensive and adaptive approach to reforms enables sustainable economic growth while minimizing social and economic risks.

Tab. 1 Summary of the reviewed literature for the part 1.1

Name of the author	Title	Methods and data used	Brief sum up
Jan Svejnar, 2002	Transition Economies: Performance and Challenges	-Comparative approach applied to selected transition economies; - Case selection: Central Europe (e.g., Poland, Hungary) vs. post-Soviet states (e.g., Russia, Ukraine); -Key variables: GDP growth, privatization patterns, institutional reforms, socio-economic indicators	Svejnar points out that Central European countries have generally performed better than their Eastern counterparts, largely due to so-called "Type II" reforms, such as improvements in legal systems and corporate governance. According to his analysis, long-term economic outcomes

			depend not only on initial conditions but also on the quality of reform strategies. He emphasizes that continued progress in legal and financial frameworks remains essential for sustained development.
Bruno Sergi, 2020	Economic Dynamics in Transitional Economies: The Four-P Governments, the EU Enlargement, and the Bruxelles Consensus	-Comparative focus on post-communist Central and Eastern Europe; -Key metrics: GDP growth, inflation, trade balance; -Analysis of privatization and market reforms; -Assessment of the EU's role in economic transformation	Sergi underscores the EU's role in shaping post-communist transitions, using the "Four-P" governance model to frame reform efforts. He argues that the "Bruxelles Consensus" is more appropriate than the Washington model. While EU-led reforms fostered growth, persistent issues—like inequality and policy gaps—call for targeted, long-term responses within the EU framework.
Aleksandr V. Gevorkyan, 2018	Transition Economies: Transformation, Development, and Society in Eastern Europe and the Former Soviet Union	-Historical study of Eastern Europe and former Soviet economies; -Indicators examined: GDP growth, inequality, poverty; -Review of institutional reforms and liberalization policies; -Social focus on migration and diaspora dynamics	Gevorkyan examines the shift from planned to market economies in post-socialist states, stressing the need to understand historical and institutional contexts. He addresses key development challenges, such as poverty, inequality, and labor migration, highlighting their continued relevance in shaping economic outcomes.
Jan Fidrmuc, 2001	Democracy in Transition	-Cross-country comparison of post-	Fidrmuc explores the link between political

	Economies: Grease or Sand in the Wheels of Growth?	communist Eastern Europe and former Soviet states; -Analysis of GDP growth, liberalization, and democracy indices; -Democracy levels assessed via Freedom House scores; -Regression used to test links between political freedom, liberalization, and economic growth	and economic liberalization in post-socialist states. He notes that democracy initially slowed growth during 1990–1993 but indirectly encouraged liberalization, which in turn supported long-term development. In his view, democracy ultimately reinforces reforms, making its overall effect on growth positive over time.
Elodie Douarin and Tomasz Mickiewicz, 2017	Economics of Institutional Change: Central and Eastern Europe Revisited ; Chapter 11: Transition as Institutional Change	-Literature-based review of institutional change in transition contexts; -Assessment of institutional models and their effects on post-communist development; -Case studies from Central and Eastern Europe used to contextualize theory	The authors explore the impact of institutional quality on economic performance during post-communist transitions. They suggest that effective market reforms require not just speed but also strong formal and informal institutions. In the absence of legal norms, property rights, and sound governance, fast liberalization may hinder lasting growth.

In order to further be able to run the comparative analysis, it is crucial to take some economic, historical and policies` context into the consideration. Therefore, the following 2 parts would be dedicated to overview of such factors for both Poland and Ukraine, with the wider focus on those steps, reforms and structural changes that were implemented in Poland, starting from the late 1980s and such of them that were ignored or implemented with the lateness in Ukraine.

1.2 Economic Transition in Poland: Path to Success

Poland successfully went through one of the most effective economic transitions among Central and Eastern European countries, driven by a combination of systemic reforms, institutional transformations, and active integration into international economic structures. The reform process began in the late 1980s, when it became obvious that the

centrally planned economy was facing deep structural issues, including low labor productivity, inefficient resource management, and chronic shortages of consumer goods (Stańczyk, 2020). In 1989, a radical reform strategy, known as the “Balcerowicz Plan”, was introduced, implementing strict macroeconomic stabilization measures and large-scale market transformations (Židek, 2011). Price liberalization was the first step toward a market economy, eliminating hard state control over pricing. This transition was accompanied by a sharp rise in inflation, which peaked at 585% in 1990, but by 1992, market-driven pricing mechanisms had stabilized the economy (Židek, 2011).

Throughout the 1980s, Polish society demonstrated a strong aspiration for economic and political change. Deep dissatisfaction with the inefficiencies of the socialist system, widespread shortages of goods, and a declining standard of living led to mass protest movements, the most significant of which was the “Solidarity” trade union. By 1980, after the waves of strikes, the government was forced to make concessions to workers, yet economic problems continued to worsen. The Polish people's desire to break away from Soviet influence and integrate into Western structures was a key factor in the determination of political elites to pursue radical reforms (Židek, 2011). The collapse of the communist regime and Poland's first free elections in 1989 demonstrated strong public support for market-oriented reforms, despite an awareness of the potential hardships associated with the transition period.

Privatization played a crucial role in Poland's economic transformation, becoming a foundation for a competitive private sector. Unlike some other post-socialist countries, Poland adopted a mixed approach to privatization, combining mass privatization, sales of assets to strategic investors, and the creation of market-based ownership mechanisms (Bellás et al., 1994). The initial stage of privatization began around 1990 and, by 1995, evolved into a more selective approach focused on selling key state assets to large-scale investors. Foreign capital played a crucial role in this transformation, contributing to industrial modernization, improved corporate governance, and the adoption of new technologies. During the early reform period, a substantial share of external investment was channeled into purchasing former state-owned enterprises, which helped speed up their restructuring and boosted their competitiveness on the global stage (Aleksander Surdej, 2000).

The reforms also had a profound impact on small and medium-sized enterprises (SMEs), which became an essential driver of employment and innovation. In 1994, a large-scale liberalization of business activity was implemented, leading to a sharp increase in the number of registered private companies. The establishment of a business-friendly environment facilitated sustained growth in the private sector, while a significant share of SMEs contributed to economic diversification (Balcerowicz, 2022). Government support measures included tax incentives, simplified administrative procedures, and access to credit resources, which accelerated business development and job creation.

Institutional reforms and Poland's integration into the European Union were key factors in the country's economic success. They and democracy were tightly interconnected in Poland's transition (Jackson, Klich & Poznańska, 2002). In 1997, a new Constitution was adopted, strengthening the legal foundations of the market economy. EU membership provided access to financial resources, simplified trade, and facilitated improvements in the legislative framework. According to Hübner (2004), integration with the EU contributed to accelerated economic growth and macroeconomic stability. So when in 2004 Poland officially became an EU member, it provided an additional boost

to economic development and attracted substantial European funding for infrastructure and industrial modernization. European standards of governance and economic transparency had an objectively positive impact on the country's business climate.

The socio-economic changes accompanying the reforms led to a significant improvement in living standards, a reduction in poverty, and better working conditions. According to Nowiak (2012), social policy reforms played a key role in Poland's successful transformation. However, the initial phase of transition was accompanied by social costs, including a rise in unemployment, which reached 14.4% in 1993, and a decline in real incomes. Nonetheless, by the late 1990s, poverty rates had steadily declined, and economic growth helped mitigate the initial difficulties (Židek, 2011).

Poland's economic transition stands as an example of a successful combination of structural reforms, macroeconomic stabilization, institutional transformation, and integration into international markets. A comprehensive and systematic approach to economic reforms enabled the country not only to overcome the crisis but also to lay the foundation for long-term sustainable development and to further enhance global competitiveness.

Tab. 2 Summary of the reviewed literature for the part 1.2

Name of the author	Title	Methods and data used	Brief sum up
Stańczyk, Z., 2020	The Success of the Polish Economic Transition	-Historical and qualitative analysis of Poland's economic transition post-1989; -Review of economic policies; -Comparative evaluation with other Central and Eastern European countries	The study examines Poland's post-socialist transition, highlighting reforms that facilitated the shift to a market economy. It identifies rapid liberalization, enterprise restructuring, and foreign investment as central factors, while also stressing the role of EU support, consistent planning, and strategic leadership in maintaining long-term growth.
Balcerowicz, L., 2022	Poland: Economic Reforms Under Extraordinary and Normal Politics—An Insider's View	-Historical overview of Poland's post-socialist transition; -Comparative analysis of policy choices across political phases; -Qualitative insights drawn from the author's experience	Balcerowicz analyzes Poland's post-socialist transformation, centering on two phases of his time as finance minister: the "extraordinary politics" of 1989–1991 and the "normal politics" of 1997–2000. He

		in economic leadership	emphasizes that reform progress under political uncertainty depended on strong leadership and a cohesive policy team.
Libor Židek, 2011	Transformation in Poland	-Historical-comparative study of Poland's late-1980s and early-1990s transition; - Review of key indicators: GDP, inflation, employment; - Analysis of Poland's shock therapy in contrast with other regional reform paths	Židek questions the notion of Poland as a unique case, arguing that its post-socialist challenges reflected broader regional trends. He explains how both domestic and international pressures drove rapid reforms—such as shock therapy—aimed at stabilization. Despite short-term costs, these steps proved vital for long-term recovery and global integration.
Danuta Hübner, 2004	Impact of the Membership in the European Union on Economic Growth in Poland	-Empirical analysis using AMECO/ECFIN data to assess GDP per capita, investment, and integration effects; - Comparative review of Poland vs. EU-15 economic outcomes; -Theoretical grounding of EU-driven convergence	The author argues that EU membership boosted Poland's growth by expanding investment, trade, and access to structural funds. However, challenges remain in closing the development gap with older EU members. Continued progress depends on broader EU support for reform and competitiveness.
Carl J. Bellas, Zbigniew Bochniarz, Władysław W. Jermakowicz, Michał Meller, David Toft, 1994	Foreign Privatization in Poland	-Empirical analysis of FDI's role in Poland's privatization; -Case studies of foreign investment in key sectors; -Statistical review of FDI flows, sectoral patterns, and employment effects	The study examines the role of FDI in Poland's post-1989 transition, emphasizing its contribution to privatization and industrial modernization, especially in the automotive and manufacturing sectors.

			While legal and administrative hurdles existed, FDI encouraged technology transfer and job growth. Initial privatization, however, was slowed by uneven reforms and limited foreign involvement.
Aleksander Surdej, 2000	Small- and Medium-Sized Enterprise Development in Poland after 1990	-Macro-level analysis of political and economic factors affecting Polish SMEs; - Statistical review of SME size, employment, and sectoral role; -Case-based assessment of SME development and obstacles	The study looks at SME development in Poland after 1990, highlighting their role in easing unemployment and maintaining social stability amid the collapse of state-owned firms. However, SMEs faced challenges such as weak innovation, low export capacity, and limited technology use. Their growth, tied to systemic gaps, remained structurally vulnerable.

1.3 Economic development challenges in Ukraine

Ukraine has faced some of the most severe challenges in economic development among post-Soviet states. Despite its wide potential, rich natural resources, and strategically important geographic location, the country has experienced prolonged economic instability, largely due to the slow pace of reforms, oligarchic control, systemic corruption, and external political pressures. Unlike most of the Central European countries, Ukraine was slow to initiate institutional transformations, which significantly slowed down its integration into international economic structures and led to prolonged economic stagnation (Minakov, 2023). Which eventually, according to Dolińska (2019), led to widespread social disparities and migration.

Throughout its independence, Ukrainian society has demonstrated highly polarized attitudes toward economic reforms. In the western regions, the population largely supported market-oriented transformations and sought integration with the European Union, viewing the EU as a guarantor of economic stability and democracy. Conversely, the eastern and southern regions, historically aligned with Russia in economic cooperation, exhibited more cautious attitudes toward market reforms and retained a certain degree of “nostalgia” for the planned economy (Kasianov, 2023). This divide in economic preferences and political orientations created additional challenges for the government, as the absence of social consensus complicated the implementation of consistent reforms. Large-scale protests, from the Orange Revolution in 2004 to

Euromaidan in 2013–2014, reflected a high level of civic engagement in economic and political transformation processes. The demand to get rid of corruption, strengthen the rule of law, and implement European economic standards became a significant factor in public pressure on the authorities, demanding accelerated reform efforts (Kasianov, 2023). But though the following the structural reforms implemented in Ukraine after 2014 have contributed to macroeconomic stabilization they faced significant implementation challenges (Ari & Pula, 2021).

One of the key challenges in Ukraine's economic transition was privatization, which took on an extremely controversial character. While in other post-socialist countries privatization facilitated the creation of a competitive private sector, in Ukraine, the process resulted in the concentration of national assets in the hands of a small group of oligarchs. Many state-owned enterprises were sold at significantly undervalued prices to business figures closely connected to the political elite, leading to the formation of powerful financial-industrial groups that made pretty obvious influence on political processes (Minakov, 2023). As a result, oligarchs gained control over key economic sectors, including energy, metallurgy, and media, which hindered the establishment of a free market and created serious barriers to foreign investment (Lerman, 1999).

Another significant challenge for Ukraine was the lack of effective measures to combat economic crime. Corruption and inefficiencies in public administration contributed to widespread financial misconduct, such as tax evasion, misappropriation of state assets, and fraud. According to the World Bank, from 2007 to 2011, corruption cost Ukraine an amount equivalent to 10% of its GDP annually, which severely undermined the country's economic sustainability (Dolbnieva, 2019).

Political instability also played a crucial role in delaying Ukraine's economic development. Ukraine retained elements of the Soviet political system for an extended period, slowing down democratization and institutional reforms (Kasianov, 2023). The absence of strong constitutional governance hindered Ukraine's early economic reforms (Kampo, 2018). One of the most significant obstacles was the lack of lustration—the process of purging former communist officials from positions of power only began 24 years after independence, which allowed authoritarian governance methods to persist and obstructed deep-rooted reforms (Kasianov, 2023). The country also experienced regional differences in attitudes towards market reforms: while the western regions strongly favored EU integration, the eastern part maintained pro-Soviet sentiments, creating political contradictions that further held back economic transformation.

Additional factor explaining Ukraine's economic difficulties was external pressure, primarily from Russia. Ukraine's dependence on Russian gas was frequently used as a political tool, significantly affecting the country's energy security. These tensions culminated in the 2013 decision to interrupt EU integration in favor of closer ties with Russia, triggering protests known as Euromaidan and leading to further instability (Minakov & Rojansky, 2015). The aftermath of these events included the annexation of Crimea, the conflict in Donbas, and the full-scale war in 2022, all of which further deepened Ukraine's economic crisis.

Despite having considerable economic potential after the fall of the Soviet Union, Ukraine has struggled for decades with serious structural issues that have impeded its growth. Oligarchic influence, corruption, the lack of consistent economic reforms, and persistent political instability have been the primary causes of economic stagnation. However, considering even present-day difficulties, the international community continues to advocate for comprehensive reforms aimed at strengthening economic

security, combating corruption, reinforcing institutional independence, and diversifying the economy (Adarov et al., 2015).

Tab. 3 Summary of the reviewed literature for the part 1.3

Name of the author	Title	Methods and data used	Brief sum up
Minakov, M., 2023	War, De-oligarchization, and the Possibility of Anti-Patronal Transformation in Ukraine	-Qualitative review of Ukraine's political economy amid and after the Russian invasion; - Assessment of de-oligarchization and its effects on governance; -Comparative analysis with post-communist transition experiences	Minakov contends that the war hastened Ukraine's political and economic restructuring, chiefly by undermining oligarchic influence. Reforms like decentralization and anti-corruption efforts, previously stalled by vested interests, began to advance. Still, efforts to curb oligarchy also pose risks—particularly the danger of authoritarian tendencies if not properly balanced.
Amat Adarov, Vasily Astrov, Peter Havlik, Gábor Hunya, Michael Landesmann, and Leon Podkaminer, 2015	How to Stabilize the Economy of Ukraine	-Macroeconomic analysis of Ukraine's current crisis using key indicators; -Comparative evaluation of Ukraine vs. other post-socialist economies; - Policy review focused on responses to economic instability	The study examines Ukraine's economic decline, shaped by reduced investment, trade disruption, and currency depreciation. Comparing it to Poland's transition, the authors highlight Ukraine's specific challenges, including entrenched oligarchy and active conflict. They advocate for robust international assistance, proposing a "Marshall Plan"-type initiative to support

			reform and postwar recovery.
Zvi Lerman, 1999	Land Reform and Farm Restructuring in Ukraine	-Empirical analysis of land ownership and farm structure in Ukraine; -Survey-based data from ~2,000 participants: private farmers, large-farm managers, and workers; -Review of legal reforms affecting land rights, collective ownership, and private agriculture	The study concludes that although legal provisions allowed for land privatization, implementation lagged, leaving much land under state ownership. Limits on land sales and leases slowed market formation. While household-level privatization had some success, wider reform was obstructed by bureaucracy, restricted farm autonomy, and little structural change—resulting in sectoral stagnation by the late 1990s.
Denisa Vadymivna Dolbnieva, 2019	Ways of Transformation of the System of Combating Economic Crimes in Ukraine Considering EU Experience	-Comparative analysis of anti-economic crime strategies in Ukraine and the EU; -Review of financial investigation models and their local applicability; -Case study of Ukraine's law enforcement and fiscal oversight systems	The study advocates for reforming Ukraine's approach to economic crime, drawing on EU models. It identifies ongoing problems such as corporate fraud, misappropriation of state assets, and major financial schemes. The authors suggest creating a financial investigation body akin to Italy's Guardia di Finanza, merging fiscal control with enforcement to enhance economic stability.

Georgiy Kasianov, 2023	In Search of Lost Time? Decommunization in Ukraine, 2014–2020	-Historical review of Ukraine's decommunization policies (2014–2020); -Policy analysis of legislative efforts to remove communist symbols; -Case studies exploring regional variation in implementation	The study underscores regional variation in the application of decommunization laws, noting broader acceptance in Western Ukraine and stronger resistance in the East. Kasianov maintains that decommunization goes beyond removing symbols, reflecting deeper tensions around national identity during political transition.
Mikhail Minakov & Matthew Rojansky, 2015	Democracy in Ukraine: Are We There Yet?	-Historical and qualitative study of Ukraine's post-independence political shifts; -Focus on political cycles, constitutional changes, and oligarch–state relations; -Analysis of elections and case studies on international and civic influence	The authors analyze how oligarchic competition, Russian influence, and decentralization affected Ukraine's political stability. They view the Orange Revolution and Euromaidan as pivotal moments that prompted short-term reforms but stress that deep-rooted corruption remains a key obstacle to long-term democratization. The study also emphasizes EU and U.S. support for reform efforts.

The economic developments in Poland and Ukraine illustrate the clear distinction in their approaches to reforms. From the early years of the post-socialist period, Poland pursued a strategy of rapid transformation, implementing decisive market-oriented reforms. Ukraine, by contrast, faced a series of institutional and structural challenges that significantly slowed its transition to a market economy. However, understanding the full extent of these differences and their long-term implications requires more than just examining each country in isolation. A direct comparison of key macroeconomic indicators, the effectiveness of institutional reforms, and the impact of external factors will help identify the foundations of Poland's success and the persistent barriers that continue to hinder Ukraine's economic growth.

1.4 Comparative Economic Studies Between Poland and Ukraine

Following the fall of the Soviet Union, Poland and Ukraine initially shared similar macroeconomic conditions. However, differences in political structures and the degree of economic autonomy played a decisive role in shaping their development trajectories. According to Wiśła and Nowosad (2020), Poland's early transition was facilitated by a combination of policy decisions and external support, including EU guidance and financial aid programs. In contrast, Ukraine, as a key republic within the USSR, was far more deeply embedded in the centrally planned Soviet economic system, which significantly complicated its transition to a market-based economy (Gylfason, Hochreiter, Kowalski, 2022). This fundamental difference explains why Part 1.2 primarily focuses on Poland's reforms during the 1990s and early 2000s, as these were the defining years of its economic transformation. Meanwhile, Part 1.3 extends Ukraine's analysis over a longer timeframe due to the prolonged and ongoing nature of its institutional and economic reforms.

A comparative look at macroeconomic trends reveals a clear gap in GDP per capita growth between Poland and Ukraine. While Poland has steadily expanded its economy, Ukraine's progress has been much more uneven, reflecting deeper structural challenges. This disproportion highlights the crucial role that strong institutions and effective governance play in fostering economic stability and growth, as discussed in Section 1.2. Poland's commitment to market reforms and policy consistency allowed it to build a more resilient economy, whereas Ukraine's prolonged political turbulence and delayed institutional changes, as explored in Section 1.3, have continued to slow down its economic development (Sliusarchuk et al., 2021). As for the decentralization, the comparative study by Savastieieva et al. (2019) highlights similarities and differences in the implementation of decentralization in both countries but concludes that effective decentralization depends on a combination of political, administrative, and financial reforms.

According to the World Bank Group (2020), Poland's accession to the European Union increased significantly the foreign direct investment (FDI), which played a crucial role in modernizing its industrial sector. Conversely, Ukraine's investment landscape was heavily dominated by capital flows through offshore jurisdictions, reducing their measurable impact on the real economy. These findings reinforce the discussion in Section 1.3, which highlights the role of weak institutions in constraining economic growth. Additionally, Wiśła and Nowosad (2020) emphasize that Poland's integration into European production networks notably boosted its export potential, whereas Ukraine, despite signing an Association Agreement with the EU in 2014, continues to face considerable challenges in fully integrating into global markets. In addition, Burdiak (2014) notes that Poland acted as a mediator between Ukraine and the European Union during critical periods and even moreover, according to Furgacz (2020) Poland adopted a more proactive stance towards Ukraine after the events of 2014.

Differences in fiscal policies between the two countries are also well-documented. Research by Sliusarchuk et al. (2021) reveals that Poland complied with strict fiscal discipline and maintained a relatively stable fiscal policy, ensuring macroeconomic stability even during economic downturns. In contrast, Ukraine frequently exceeded sustainable deficit levels, with a significant portion of its public debt allocated to short-

term expenditures rather than long-term economic development. This pattern aligns with the conclusions of Section 1.3, which highlights Ukraine's recurring financial instability and its broader economic consequences.

Institutional governance also plays a crucial role in shaping economic performance. As noted by Wisła and Nowosad (2020), Poland achieved significant progress in reducing corruption, fostering a more transparent business environment, and attracting higher levels of foreign investment. In Ukraine, however, high corruption levels remain a major obstacle to economic progress. This persistent issue is reflected in various global governance indicators further examined in existing research.

Another decisive factor in economic development is innovation. Sliusarchuk et al. (2021) point out that Poland has actively promoted high-tech industries through tax incentives and other supportive measures. In contrast, Ukraine remains heavily reliant on raw material exports, limiting its ability to foster an innovation-driven economy. These observations further confirms Section 1.3's assessment of weak innovation policies and their impact on economic competitiveness.

The overall analysis supports the idea that Poland's economic success can largely be attributed to systematic reforms, EU membership, improved institutional governance, and an active push for innovation. In contrast, Ukraine continues to struggle with deep-rooted challenges, including corruption, inefficient economic policies, and weak integration into global markets. These inconsistencies underscore the pressing need for further institutional reforms in Ukraine to strengthen economic resilience and create an environment conducive to sustainable growth.

Tab. 4 Summary of the reviewed literature for the part 1.4

Name of the author	Title	Methods and data used	Brief sum up
Pyatnytska G. T., Kucherenko D. O., Pavlenko I. S., 2020	Key Success Factors in Domestic Trade: The Experience of Ukraine and Poland	-Comparative study of domestic trade in Ukraine and Poland; -Analysis of regulatory frameworks and retail structures; - Statistical data from national offices in both countries	The study compares domestic trade in Ukraine and Poland, observing that small enterprises prevail in both, though they are more concentrated in Poland, particularly in food retail. Supportive tax policies and SME incentives in Poland encouraged investment stability, while Ukraine continues to face weaker business concentration and less supportive regulatory conditions.
World Bank Group, 2020	Doing Business Comparing Business	-Quantitative comparison of business	The study contrasts Poland and Ukraine across key regulatory

	Regulation in 190 Economies 2020	environments in 190 countries; - Benchmarking through global rankings; - Standardized case studies for cross-country consistency	domains such as business registration, construction permitting, and credit access. Poland performs better thanks to stronger institutions and simpler procedures, whereas Ukraine's regulatory complexity and limited investor safeguards pose major obstacles to business operations.
Sliusarchuk, O., Lavrov, R., Kuybida, V., Slatvinskyi, M., & Zelenskyi, A., 2021	Fiscal Policy of Economic Development: Comparative Characteristics of Ukraine and Poland	-Statistical comparison of fiscal policy indicators in Ukraine and Poland; -Correlation analysis of fiscal variables and economic growth; -Economic modeling to assess fiscal policy effects on GDP	The authors emphasize Poland's strategic use of EU funding and infrastructure co-financing, which fostered robust GDP growth and modernization. Conversely, Ukraine's fiscal policy depends on deficit spending and lacks stable debt management, hindering outcomes. Poland's sound public debt strategy aided growth, while Ukraine remains limited by external debt and weak fiscal oversight.
Rafał Wiśła, Andrzej Nowosad, 2020	Economic Transformation in Poland and Ukraine: National and Regional Perspectives	-Comparative analysis of economic transformation in Poland and Ukraine; -Study of regional policy impacts on local development; -Qualitative review of historical events shaping growth dynamics	The study credits Poland's success to early reforms, EU accession, and efficient use of development funding. In contrast, Ukraine encountered political instability, armed conflict, and inconsistent policies that hampered growth. It also underscores regional disparities in both countries, reflecting uneven patterns of

			economic development across their territories.
Thorvaldur Gylfason, Eduard Hochreiter, Tadeusz Kowalski, 2022	Different Choices, Divergent Paths: Poland and Ukraine	-Comparative analysis of growth paths in Poland and Ukraine using economic and social indicators; - Mixed-method evaluation of institutions, trade, governance, democracy, and investment factors	The study concludes that Poland's swift transformation was fueled by EU-oriented reforms, effective governance, reduced corruption, and targeted investment in education and industry. In contrast, Ukraine faced stalled reforms, political divisions, and entrenched oligarchy. Consequently, by 2021, Poland's GDP per capita nearly tripled that of Ukraine.

1.5 Partial Conclusions Based on Literature Review

The literature review has identified key factors influencing the success of economic transition in post-socialist countries, using Poland and Ukraine as case studies. The transformation from a centrally planned economy to a market-based system posed significant challenges, including macroeconomic stabilization, privatization, institutional reforms, and integration into international economic structures. However, despite sharing common transition goals, Poland and Ukraine adopted different approaches, which ultimately led to contrasting economic outcomes.

First and foremost, one of the main factors determining the differences in their economic development was the strategy for implementing reforms. Poland pursued a radical "shock therapy" approach, allowing for a rapid adjustment to new market conditions but at the cost of short-term socio-economic disruptions. Price liberalization, reduced government intervention, and accelerated privatization triggered a temporary surge in inflation. However, in the long run, these measures facilitated economic growth. Ukraine, on the other hand, opted for a more gradual transition, characterized by frequent shifts in economic policy direction. While this approach helped soften the initial shock, it also slowed down the formation of market mechanisms and created uncertainty in the business environment.

Secondly, the nature and mechanisms of privatization played a crucial role in shaping the economic trajectories of these countries. Poland implemented a comprehensive approach that included mass voucher privatization, asset sales to strategic investors, and policies aimed at fostering entrepreneurship. As a result, a competitive private sector emerged, and foreign direct investment (FDI) increased. In contrast, Ukraine's privatization process was plagued by corruption and a lack of transparency, leading to the concentration of assets in the hands of a narrow elite. This, in turn, hindered competition and reduced the country's attractiveness to investors.

Thirdly, institutional reforms and political stability significantly influenced economic development. Poland focused on strengthening its legal framework, protecting property rights, improving corporate governance, and implementing anti-corruption measures. As a result, the country's political stability encouraged investment inflows and created a favorable business environment. Ukraine, however, struggled with weak institutions, high levels of corruption, and an unstable political course. The frequent shifts in economic strategy, without a clear long-term vision, further complicated the transition process.

Another key factor in the economic transformation of these countries was the role of the European Union. Poland's commitment to EU integration secured substantial financial aid, investment, and expert support for reforms. EU accession granted Polish businesses access to new markets, stimulated export growth, and enhanced the country's overall competitiveness. Conversely, Ukraine remained under Russia's sphere of influence for an extended period, limiting its ability to implement large-scale economic reforms. The slow progress in aligning with EU policies and the absence of a coherent modernization strategy further hindered Ukraine's economic development.

Moreover, the social consequences of reforms cannot be underestimated. In Poland, radical reforms initially led to a temporary rise in unemployment and a decline in real incomes. However, in the long run, these changes contributed to an overall improvement in living standards. In Ukraine, the prolonged nature of reforms led to social challenges, expanded the informal economy, and led to a significant outflow of skilled professionals abroad. These trends negatively impacted Ukraine's long-term economic growth potential.

Thus, the findings of the literature review suggest that the success of economic transition is not solely determined by macroeconomic factors but is also heavily influenced by the quality of institutional reforms, the degree of political stability, and the strategic direction of economic policies. Poland's experience demonstrates that consistent reforms, international support, and long-term planning can lead to sustainable economic growth. Ukraine, however, continues to face obstacles related to inconsistent reforms, corruption, and political instability. Nonetheless, several positive factors—such as deeper trade relations with the EU and gradual reforms in the public sector—offer prospects for future economic development.

These conclusions lead into the next part of the study, which will focus on analyzing how developed the economies are, how institutions function, and how effective economic policies are compared to other countries in the region. The next section takes a closer look at key macroeconomic indicators, the quality of governance, and recent trends in trade and investment. This analysis will allow for a deeper understanding of key economic challenges and growth prospects while also offering the foundation for policy recommendations for improving economic strategies within the framework of European integration.

2 Methods and data

In order to understand the reasons behind the economic gap between Ukraine and Poland since the early 1990s, this thesis uses a comparative, data-oriented approach. The idea was not only to track changes in key macroeconomic indicators, but also to place them within a broader context — one that includes political shifts, institutional development, and the way each country integrated into the global economy. The method combines time series econometrics, descriptive analysis, and contextual interpretation. This mix allows for both country-level comparisons and a closer look at how certain key events influenced economic performance over time.

Three main groups of indicators were used to support the empirical part of the research. The first group includes GDP per capita, inflation rate, and unemployment rate — traditional indicators that reflect the overall economic health of a country. They were chosen because they give a good sense of macroeconomic trends and help capture the longer-term direction of each economy.

The second set of indicators is more focused on governance and institutional quality. For this, data from the Corruption Perception Index (CPI) by Transparency International and several dimensions from the World Bank's Worldwide Governance Indicators (WGI) were used — specifically, Control of Corruption, Government Effectiveness, and Regulatory Quality. These indicators are based on perception and expert surveys, but they are still widely used in both academic research and policymaking, especially when it comes to tracking institutional change.

The third group of variables looks at how open each country is to the global economy. This includes trade balance and foreign direct investment (FDI) inflows, both measured as a percentage of GDP. For all of the mentioned indicators, time series regressions were run using the Gretl software. Structural breaks were tested with the help of Chow tests and QLR tests, and dummy variables were introduced in the models where needed — to capture changes in trends around major events like EU accession, political revolutions, or the start of the war in Ukraine.

In addition to the core econometric analysis, the thesis also incorporates more descriptive comparisons. First, the export structure and key trade partners of both countries were analyzed for the years 1995, 2010, and 2023. These years were selected to reflect the beginning of transition, a midpoint of reform, and the most recent available data. The export analysis focuses not only on what each country exports, but also on where those goods go. This gives additional insight into whether the countries moved toward more complex and value-added export structures — or remained stuck in commodity cycles. The data for this part was taken from the OEC (Observatory of Economic Complexity).

Second, a breakdown of FDI inflows by economic sector was added, based on official data for 2010 and 2018 — the only two years with usable, comparable data for both countries. This part helps clarify which sectors attracted foreign capital and whether there has been any significant shift toward more productive or technology-intensive areas.

The empirical econometric analysis was conducted in Gretl, using Ordinary Least Squares (OLS) regressions. In each model, structural breaks were tested using both Chow and Quandt Likelihood Ratio (QLR) tests. Dummy variables and time interactions were

added where appropriate to reflect periods of reform, external shocks, or institutional changes. The aim was not to build predictive models, but to detect meaningful trend shifts and connect them to real-world events.

Originally, it was planned to include data starting from the late 1980s to capture the earliest stages of economic transition in both countries. However, after extensive searching, it turned out that reliable and comparable datasets for both Ukraine and Poland are only consistently available starting from the early 1990s. For this reason, the analysis covers the period from the early 1990s up to 2023, depending on the availability of each specific indicator. The data used comes primarily from recognized sources such as the World Bank, Transparency International, Macrotrends, and national statistical offices. To make comparisons meaningful across indicators and countries, most values were adjusted — for example, expressed in constant prices, indexed, or calculated as a share of GDP.

The table below provides a summary of all the main indicators used in the research, along with the time periods they cover, the way they are measured, and the sources they were taken from.

Tab. 5 Summary of data used in the research

Indicator	Period	Value expression	Data source
GDP per capita	1990-2023	Current US \$, constant 2015 prices	Worldbank
Inflation	1993-2023	% change in consumer prices	Worldbank
Unemployment	1991-2021	% of labor force, national estimate	Worldbank
Corruption perception index	2012-2024	Index (0= highly corrupt; 100=clean)	Transparency International
Control of corruption (WGI)	1996-2023	Index (-2,5 to +2,5)	Worldbank
Government Effectiveness (WGI)	1996-2023	Index (-2,5 to +2,5)	Worldbank
Regulatory Quality (WGI)	1996-2023	Index (-2,5 to +2,5)	Worldbank
Trade Balance	1995-2023	% of GDP	Macrotrends
FDI inflows	1992-2023	% of GDP	Worldbank

3 Empirical analysis

This part of the thesis takes a closer look at how the economies of Poland and Ukraine have evolved over time by examining real data across different areas. Instead of focusing on abstract theories, the goal here is to work directly with numbers — looking at trends, shifts, and patterns that shaped both countries' development. The analysis is based on time series models built in Gretl, with the length of each dataset depending on how far back reliable data goes. For clarity and consistency, the indicators are organized into three main categories: overall economic performance, institutional and governance quality, and trends in trade and investment.

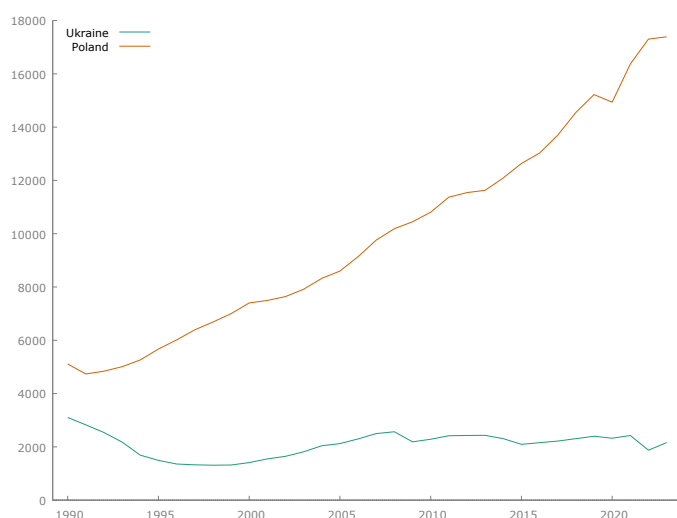
3.1 Overview of Basic Economic Indicators

To better understand how economic development evolved in Ukraine and Poland—and to compare the speed and direction of their transformation—it makes sense to start by looking at key macroeconomic indicators. This part of the study focuses on three main ones: GDP per capita, inflation, and unemployment rates. These indicators were chosen because each highlights a different aspect of a country's economic situation.

GDP per capita reflects how much output is produced per person and gives a general idea of the population's average standard of living. The inflation rate reflects the stability of prices and the economy's ability to maintain monetary balance — which is especially relevant for countries that went through a turbulent transition period. Meanwhile, the unemployment rate provides a snapshot of the labor market and shows how actively the population is engaged in economic activity. When analyzed together, these indicators offer a comprehensive view that is crucial for understanding both the differences and the parallels in the economic trajectories of Ukraine and Poland — particularly in light of the macroeconomic shifts that have taken place since the 1980s.

To begin, let us take a closer look at GDP per capita and how it has changed for both countries from 1990 to 2023.

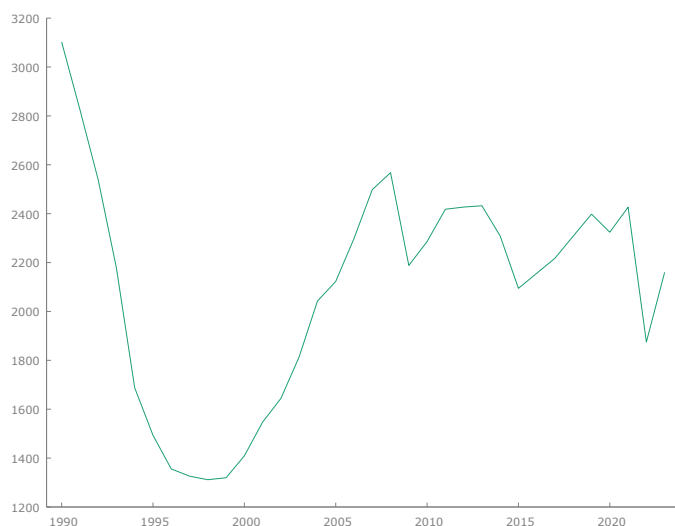
Fig. 1 Development of GDP per capita in Ukraine and Poland during the period 1990 – 2023, expressed in constant 2015 prices



At the very beginning of the analysis, it becomes clear that Poland is a significantly wealthier country compared to Ukraine, at least when looking at GDP per capita. Between 1990 and 2023, Poland's average GDP per capita amounted to \$9,888.21 (in constant 2015 prices), while for Ukraine, the figure was just \$2,091.30. The chart clearly shows that Poland experienced nearly uninterrupted growth throughout the entire period, with a

particularly sharp acceleration after 2004. Ukraine, on the other hand, went through a deep economic decline in the 1990s and only managed a partial recovery in the 2000s, consistently remaining almost five times behind Poland in terms of average GDP per capita.

Fig. 2 GDP per capita of Ukraine for the period 1990 – 2023, expressed in constant 2015 prices

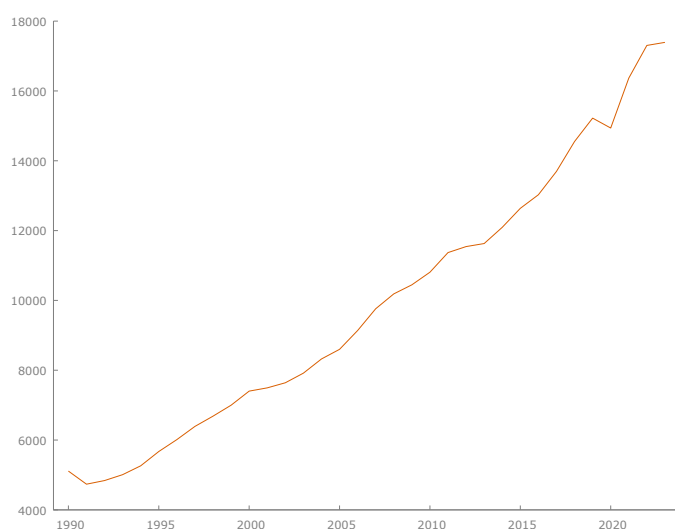


The chart reveals a U-shaped dynamic in Ukraine's GDP per capita: after a deep decline in the 1990s, growth began to recover in the early 2000s. Despite this upward shift, the series remains volatile, especially during the early post-Soviet transition.

To detect potential structural shifts, both QLR and Chow tests were applied to the OLS model. The results confirmed a statistically significant break in the year 2000 — affecting both the level and the slope of the

trend. This reflects not only a rise in average GDP per capita but also a broader shift from economic decline to steady growth. The turning point followed a reorientation of Ukraine's economic policy: the government launched structural reforms, stabilized macroeconomic indicators, and boosted metal and chemical exports. The tax system also became more regulated, including the adoption of a fixed agricultural tax (Havrylyshyn et al, 2002). Still, despite these initial improvements, Ukraine's economic course remains fragile. Repeated political and economic disruptions, especially in 2014 and 2022, have continued to limit its long-term growth prospects.

Fig 3. GDP per capita of Poland for the period 1990 – 2023, expressed in constant 2015 prices



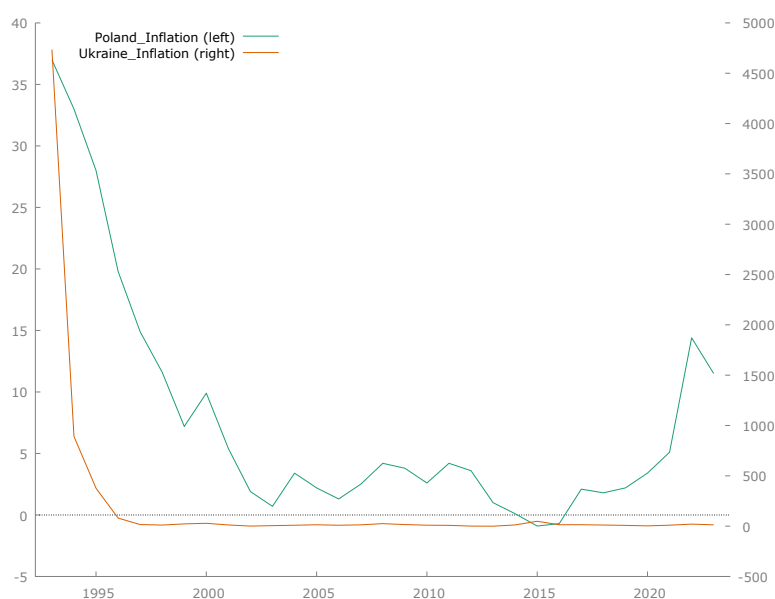
Poland's GDP per capita from 1990 to 2023 shows a steady upward trend, with only minor fluctuations. This reflects consistent macroeconomic growth and stable long-term development.

To identify potential structural shifts, QLR and Chow tests were applied to the OLS model. Both confirmed a statistically significant break in 2014. The negative coefficient for the level (break2014) suggests a temporary decline, while the

positive slope (break2014t) indicates accelerated growth afterward. This decline likely resulted from external shocks, including regional geopolitical tensions, yet Poland rebounded quickly and resumed a strong growth trajectory. From 2014 onward, the country actively drew on the new EU budget for 2014–2020, which provided around €82.5 billion. Much of this funding went into infrastructure and industrial development. Growth was further supported by rising exports—especially to Germany—and by increased domestic consumption, aided by low inflation. Despite instability linked to the crisis in Ukraine, Poland outpaced the EU average in economic growth during this period (Czyżewski & Matuszczak, 2020). In general, although short-term fluctuations occurred, Poland’s GDP per capita maintained a steady upward trend, clearly differing from the more volatile dynamics seen in other post-Soviet states.

Looking at GDP per capita gives a good sense of how much an economy produces per person, but it doesn’t tell us how stable prices are or how far people’s money actually goes. That’s why it makes sense to now shift focus to inflation rate and how it’s changed over time in both countries.

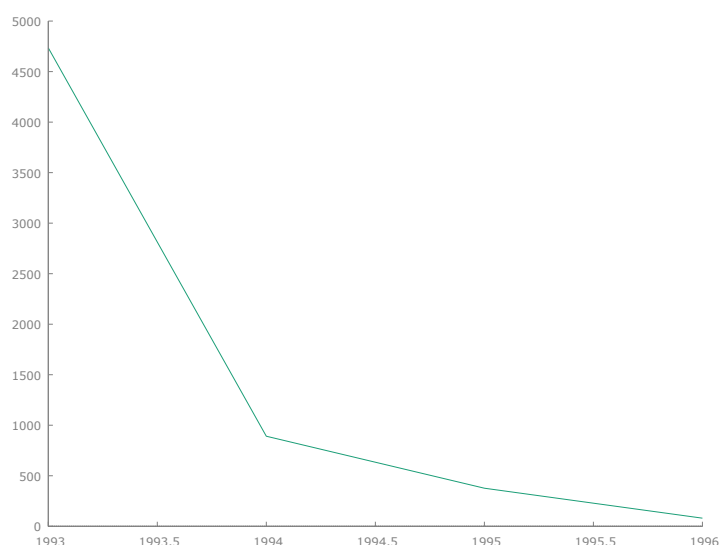
Fig. 4 Inflation in Ukraine and Poland in % annual change, for the period 1993-2023



The chart presents annual inflation trends in Ukraine and Poland from 1993 to 2023. In the early years, Ukraine faced hyperinflation—often dozens or even hundreds of times higher than in Poland, where macroeconomic stabilization was already in progress. By the early 2000s, both countries had reduced inflation to single digits, though Ukraine’s rates remained far more

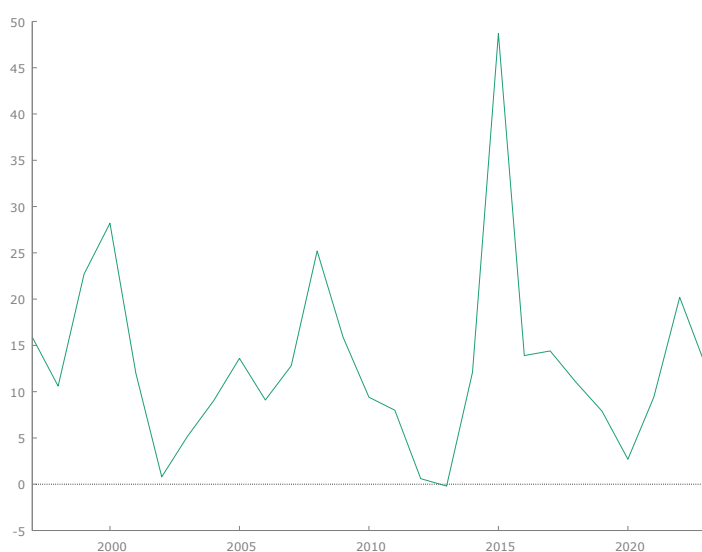
volatile, especially during major crises in 2008, 2015, and 2022. Poland, by contrast, kept inflation largely stable, with only brief spikes caused by external shocks. A dual Y-axis was used to preserve visual clarity, given the significant gap in inflation levels between the two countries. Due to the sharp contrast between early and later values, the data for Ukraine would further be splitted into two graphs (1993-1996 and 1997-2023) to maintain clarity and prevent compression of recent trends.

Fig. 5 Development of inflation rate in Ukraine for the period 1993-1996



The first inflation time series analyzed relates to Ukraine. In 1993, the country experienced extreme hyperinflation — prices surged by over 4,700%, a direct consequence of the abrupt shift from a planned to a market economy and the absence of effective monetary policy. Over the next three years, inflation dropped sharply, and by 1996, macroeconomic stabilization had begun.

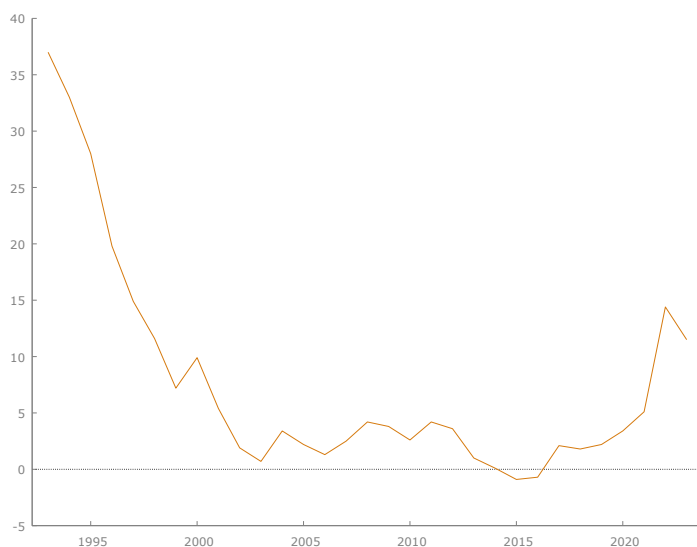
Fig. 6 Development of inflation rate in Ukraine for the period 1997-2023



From 1997 to 2023, inflation remained more moderate, with notable spikes during major crises — in 1999, 2008, 2014, 2015, and 2022. Both the QLR and Chow tests confirmed a structural break in 1996, affecting inflation's level and trajectory. As noted by Kravchuk (1998), the introduction of the hryvnia in September 1996, replacing the severely devalued karbovanets, marked a key moment in Ukraine's post-Soviet stabilization. This

reform helped bring hyperinflation under control. It was paired with legislation establishing the National Bank of Ukraine as an independent body responsible for inflation control and monetary stability. Thus, 1996 is seen as a turning point toward relative price stability.

Fig. 7 Development of inflation rate in Poland for the period 1993-2023



The next time series shows Poland's inflation patterns from 1993 to 2023. Like many transition economies, Poland faced high inflation in the early 1990s, but within ten years, price growth slowed sharply. This was largely due to the country's shock therapy program, market liberalization, structural reforms, and active central bank intervention. After 2002, inflation became more stable and stayed near

target levels, even during global disruptions like the 2008 financial crisis and the COVID-19 pandemic. QLR and Chow tests confirmed a significant structural break in 2002, affecting both the level and trend of inflation. Before that point, inflation was falling; afterward, it stabilized at lower rates. Ahead of joining the EU in 2004, Poland prioritized meeting the Maastricht criteria, focusing particularly on reducing inflation. Thanks to tight monetary policy by the National Bank of Poland, inflation dropped to around 1.9%. The government also reinforced fiscal discipline and kept a conservative stance to align with EU norms (Wojciechowski, 2014).

While inflation shows us how stable or unpredictable prices have been, it doesn't say much about whether people actually had jobs or steady incomes during those times. To get a fuller picture of economic health, the next step is to look at unemployment trends in both countries.

Fig. 8 Unemployment rates in Ukraine and Poland during the period 1991 – 2021, expressed as the % of total labour force



The chart displays unemployment patterns in Ukraine and Poland between 1991 and 2021. In the 1990s and early 2000s, Poland faced much higher unemployment, reaching its peak around 2002, largely due to extensive market reforms and economic restructuring. In the years that followed, Poland's unemployment rate dropped noticeably, particularly during its EU accession preparations.

Ukraine's unemployment increase started a bit later and remained less severe overall. Still, unlike Poland, Ukraine showed no clear downward trend, with unemployment staying relatively stable over time. On average, the unemployment rate was 11.17% in Poland and 7.86% in Ukraine during the observed period.

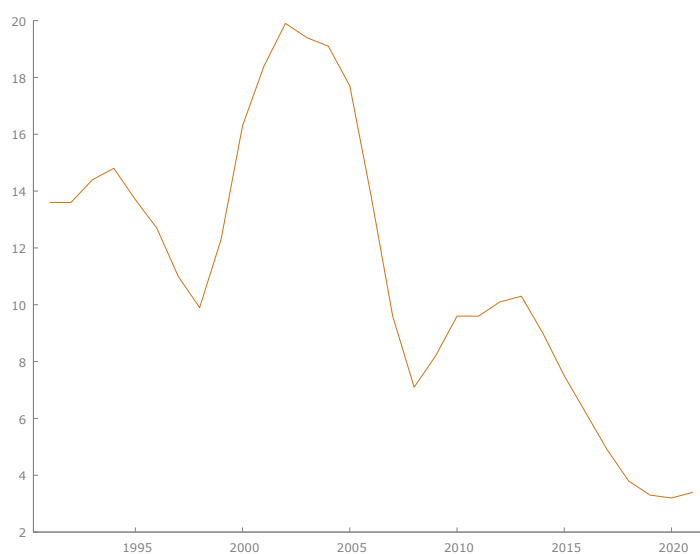
Fig. 9 Development of unemployment rate in Ukraine during the period 1991 – 2021, as a % of total labor force



Ukraine's unemployment time series shows a steady increase through the 1990s, reaching its peak in 2002. This reflects the impact of post-Soviet transition, major public sector downsizing, and a slow adjustment to a market-based labor system. Both QLR and Chow tests confirm a structural break in 2002, with a sharp rise followed by gradual improvement. As noted by Kupets (2006), the early 2000s brought key labor market reforms, especially the 2001

employment law, which took effect in 2002. This legislation expanded public employment services and emphasized worker retraining. At the same time, Ukraine's economy began to grow—GDP rose by 5.2% in 2002—helping reduce unemployment, particularly in industrial regions. This shift may also reflect increased labor migration, informal work, and fewer officially registered job seekers.

Fig. 10 Development of unemployment rate in Poland during the period 1991 – 2021, as a % of total labor force



The chart shows unemployment trends in Poland from 1991 to 2021. Rates remained high through the 1990s, peaking in the early 2000s due to market reforms, enterprise closures, and major labor restructuring. QLR and Chow tests identified a structural break in 2000: unemployment had been steadily rising before then, but began to decline afterward. As noted by Bukowski et al. (2007), joblessness was largely

driven by post-privatization slowdown and deep economic adjustments. Many small, inefficient firms—especially in agriculture and low-productivity sectors—were forced to close, adding to labor market disruption. Meanwhile, signs of economic overheating emerged as Poland accelerated preparations for EU accession. From the 2010s on, unemployment steadily declined, falling below 3% by 2021—well under the EU average. These trends underscore the flexibility of Poland’s labor market and the impact of employment policy. Interestingly, Koshulko and Koshulko (2016) argue that labor migration from Ukraine had a positive and direct effect on Poland’s workforce.

To summarize the findings from this section, the following table provides a comparative overview of key trends and structural changes in GDP per capita, inflation, and unemployment across Ukraine and Poland.

Tab. 6 Summary regarding the models of time series for the basic economic indicators

Indicator	Country	Observed period	Model	R ²	Break year	Trend shift	Conclusion
GDP per capita	Ukraine	1991 - 2023	OLS + break	0.77	2000	Shift from decline to recovery	Ukraine’s output per capita started low and fell further in the ‘90s, but a turning point around 2000 sparked a modest recovery, though still fragile.
	Poland	1991 - 2023	OLS + break	0.99	2014	Temporary dip, accelerated growth	Poland’s GDP kept climbing almost steadily; the 2014 drop didn’t last long and growth resumed quickly, likely due to strong fundamentals.
Inflation	Ukraine	1994 - 2023	OLS + break	0.997	1996	Transition to price stability	After the chaos of hyperinflation, Ukraine managed to stabilize prices post-1996, although the situation remains sensitive during crises.
	Poland	1994 - 2023	OLS + break	0.92	2002	Long-term moderation	Inflation in Poland dropped steadily after major reforms. Since the early 2000s, it has mostly stayed under control despite some global turbulence.
Unemployment	Ukraine	1991 - 2021	OLS + break	0.82	2002	Slowed increase, mild decrease	Unemployment in Ukraine rose after independence and peaked in the early 2000s. Since then, it declined a bit but stayed relatively high.
	Poland	1992 - 2021	OLS + break	0.90	2000	Sharp decline	Poland had a tough labor market in the ‘90s, but joblessness fell quickly after 2000, likely helped by EU-related reforms and investments.

3.2 Institutional quality and governance

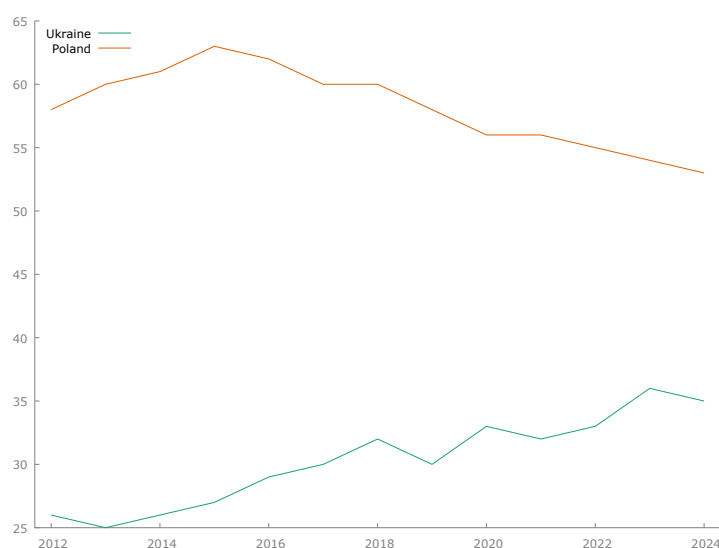
When it comes to long-term economic growth, macroeconomic stability alone isn't enough. Institutions — how the state functions, how transparent, predictable, and effective it is — play a crucial role. This is particularly true for transition economies like Ukraine and Poland, where the success of reforms largely depends on the quality of governance. Strong institutions not only improve a country's investment climate but also boost public trust, reduce corruption, and lay the groundwork for sustainable development.

This part of the study looks at how institutional conditions have influenced the economic paths of Ukraine and Poland. For the analysis, the widely accepted indicators such as Transparency International's Corruption Perceptions Index (CPI) and the World Bank's Worldwide Governance Indicators (WGI) are being used. The focus is on three core aspects of governance — corruption control, government effectiveness, and regulatory quality — and how these have changed over time. These indicators help track not only the state of institutions but also key shifts that may have affected growth, investment, and overall development.

By comparing Ukraine and Poland through the lens of institutional quality, we can better understand why their economic outcomes diverged so significantly, despite similar starting conditions in the early 1990s. While economic indicators show the effects, institutional ones often point to the underlying causes.

We begin with the Corruption Perceptions Index (CPI) by Transparency International. The Corruption Perception Index doesn't measure corruption directly, but rather reflects how corrupt a country is perceived to be by experts and businesspeople. It's a useful way to gauge public trust in institutions and the integrity of political and economic systems. A higher CPI score means lower perceived corruption — and often signals a more attractive and predictable environment for both citizens and investors.

Fig. 11 Corruption perception index development in Ukraine and Poland during the period 2012-2024



The chart illustrates both the gap in perceived corruption between Ukraine and Poland and the contrasting directions each country has followed over the past 12 years. Poland began the period with a relatively high CPI score, suggesting low corruption levels, but has seen a gradual decline since 2016. Ukraine, by contrast, started with one of the lowest scores in Europe, yet has made slow, consistent progress—particularly after 2014, likely

influenced by anti-corruption reforms and pressure from international actors. Between 2012 and 2024, Poland's average CPI score was 58.15, while Ukraine's stood at 30.31.

To better capture the contrast in trajectories and make the dynamics more visible, the Y-axis was limited to the range of 25–65, even though the official CPI scale ranges from 0 to 100.

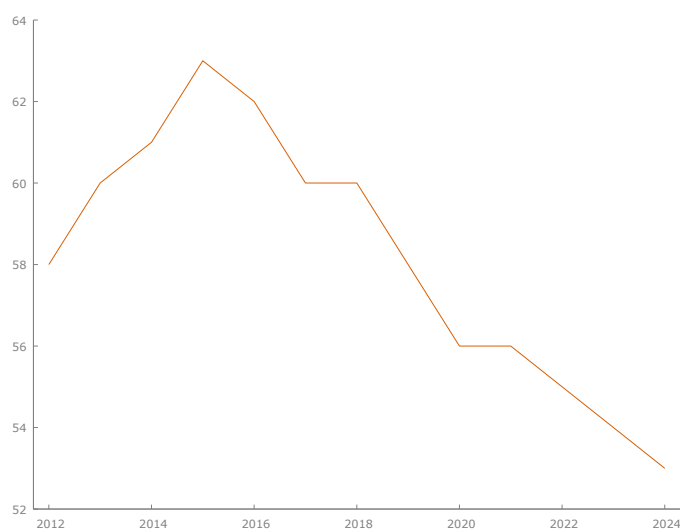
Fig. 12 Corruption perception index development in Ukraine during the period 2012-2024



The time series shows a gradual rise in Ukraine's perceived corruption score—from 26 in 2012 to 36 in 2023. This steady progress likely stems from several factors. After 2014, Ukraine ramped up anti-corruption efforts by creating specialized agencies like NABU and SAPO, and adopting digital transparency tools. International partners, including the EU and IMF, also increased pressure by tying financial assistance to rule-of-law reforms. Meanwhile, civil

society became more engaged, actively overseeing public spending and advocating institutional change. Neither Chow nor QLR tests revealed any significant structural break, indicating that improvements occurred gradually rather than abruptly. This suggests CPI gains reflect the cumulative effect of reform, though the short 12-year span of data limits broader conclusions.

Fig. 13 Corruption perception index development in Poland during the period 2012-2024



Poland's CPI trend splits clearly into two phases. From 2012 to 2015, the index rose from 58 to 63, reflecting optimism about anti-corruption efforts during that period. However, starting in 2016, the trend reversed: CPI scores began to decline, dropping to 53 by 2024. Both Chow and QLR tests confirmed a significant structural break in 2016, indicating a shift in both the level and direction of the trend. This turning point

coincides with the 2015 election victory of the Law and Justice party (PiS), after which the government introduced judicial reforms that quickly sparked criticism from the EU. In particular, limits placed on the Constitutional Tribunal raised concerns about democratic backsliding and weakening rule-of-law standards. As Balcer (2016) points

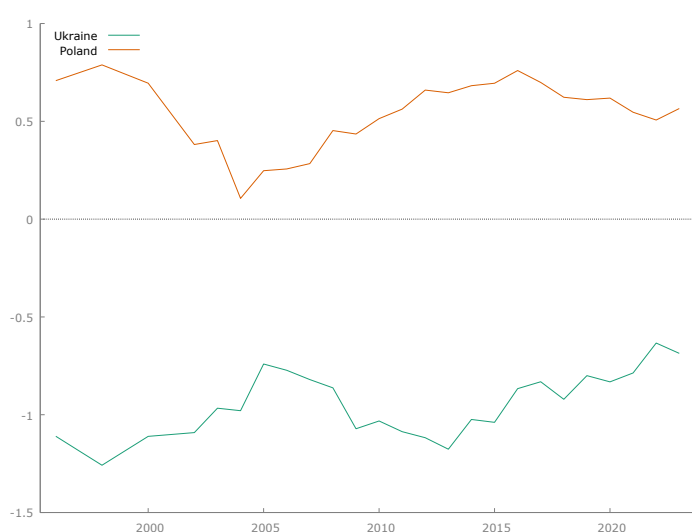
out, these changes damaged Poland's image in the EU and heightened concerns over institutional integrity and transparency. While Poland continued to score relatively high in Transparency International's CPI, the 2016 report marked the start of a downward trajectory. This case shows how anti-corruption progress can produce short-term results but may not be sustainable without strong institutions and democratic safeguards.

Before moving on to the analysis of the Worldwide Governance Indicators (WGI), it is important to clarify the approach taken to working with the available data. WGI indicators are presented in two formats — Estimate and Rank. For the purpose of building econometric models in this research, the Estimate format is used, as it is a continuous variable ranging from -2.5 to +2.5. This makes it more suitable for quantitative analysis, allowing the tracking of subtle changes and the application of standard regression methods. At the same time, visualizations such as graphs and charts include Rank values, as they reflect a country's relative position and are easier to interpret visually.

Another important aspect concerns the time structure of the data. Originally, WGI data between 1996 and 2002 were published every two years, resulting in gaps for some years. To ensure a consistent annual time series and allow for econometric analysis using yearly observations, missing values for 1997, 1999, and 2001 were interpolated. This approach helped preserve the full timeline without reducing the sample size. Interpolation does not distort the overall trend and creates a more stable basis for analyzing the relationship between governance quality and economic development.

The first observed indicator is Control of Corruption which captures how effectively a country prevents the abuse of public power for private gain — whether through bribery, patronage, or weak enforcement. It reflects not just the presence of corruption, but how well institutions are able to keep it in check. Higher scores suggest that anti-corruption efforts are working and that the rules apply more equally to everyone.

Fig. 14 Control of Corruption estimate index development in Ukraine and Poland during the period 1996-2023



The graph illustrates the evolution of the Control of Corruption indicator for Ukraine and Poland, reflecting each country's ability to effectively curb corruption. Higher values (closer to +1) indicate stronger anti-corruption institutions, while negative values (closer to -1) point to weak control and widespread corruption. To improve readability and highlight the trends in both countries, the chart uses two vertical axes. This was necessary due to the large

gap between Ukraine's and Poland's values — without separate axes, Ukraine's trajectory would appear compressed, making it difficult to analyze the changes over time. The dual-axis format allows for a more balanced and informative visual comparison.

Throughout the observed period, Poland consistently stayed in the positive range, suggesting relatively strong institutional capacity to combat corruption. Its accession to the EU in 2004 appears to have played a particularly beneficial role, supporting steady gains and eventual stabilization. Ukraine, by contrast, remained in the negative zone over the entire timeframe, reflecting deep-rooted structural obstacles in its anti-corruption efforts. Still, since 2014, and despite ongoing political and military challenges, Ukraine has shown gradual progress—likely driven by institutional reforms, international pressure, and growing public demand for transparency. On average from 1996 to 2023, Poland scored about +0.51, while Ukraine averaged -0.92 , highlighting a persistent, though gradually narrowing, gap in corruption control between the two.

Fig. 15 Control of Corruption estimate index development in Ukraine during the period 1996-2023



Ukraine's Control of Corruption time series reflects a volatile pattern shaped by recurring crises, reform efforts, and persistent institutional weaknesses. Between 1996 and 2000, the indicator stayed below -1.1 , which was typical of post-Soviet transition periods marked by instability, weak institutions, and widespread informal practices. A brief improvement occurred from 2005 to 2007, likely linked to the Orange

Revolution and expectations of democratization, but this momentum faded by 2010. Both QLR and Chow tests identified a statistically significant structural break in 2009. As Kuzio (2013) explains, this period was marked by institutional stagnation driven by growing conflict between President Yushchenko and Prime Minister Tymoshenko. Their rivalry within a fragmented democratic movement blocked key reforms, especially in the anti-corruption sphere. The lack of unified leadership undermined public trust and strengthened perceptions of worsening corruption and informal power networks. This political deadlock also led to increased pressure on businesses and more visible selective policymaking, as noted by external observers. Since 2014, however, a gradual upward trend has emerged—fueled by political shifts, EU-driven reforms, and external pressure for transparency. The positive and significant time variable in the model indicates steady improvement in corruption control over the last decade.

Fig. 16 Control of Corruption estimate index development in Poland during the period 1996-2023

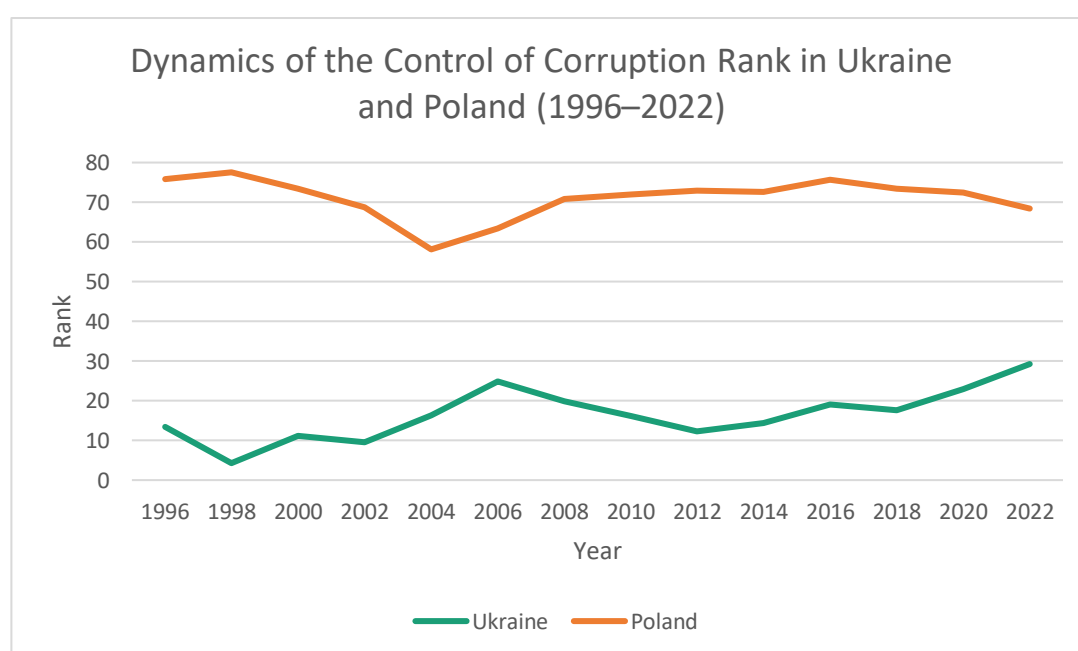


Since the late 1990s, Poland has generally maintained a relatively strong level of corruption control, though some fluctuations did occur. A dip in the early 2000s overlapped with political instability and several public scandals. From 2005 onward, the situation gradually improved, supported by EU accession, institutional reforms, and tighter oversight. Poland reached its highest score around 2015, followed by a period of moderate decline and eventual

stabilization. The regression analysis confirms a structural break in 2008, as shown by both QLR and Chow tests. Anti-corruption efforts gained traction following the establishment of the Central Anti-Corruption Bureau (CBA) in 2006—an institution created to investigate corruption in public and economic sectors (Ociecek, 2023). By 2008, the CBA had increased its visibility through several high-profile cases. Although the model shows a post-2008 improvement, the negative time coefficient points to a gradual weakening of long-term momentum.

To complement the regression analysis, the chart below illustrates the dynamics of the Control of Corruption ranks in Ukraine and Poland from 1996 to 2022. This ranking-based perspective offers additional insight into how both countries have positioned themselves globally over time in terms of institutional integrity.

Fig.17 Dynamics of the Control of Corruption Rank in Ukraine and Poland 1996–2022

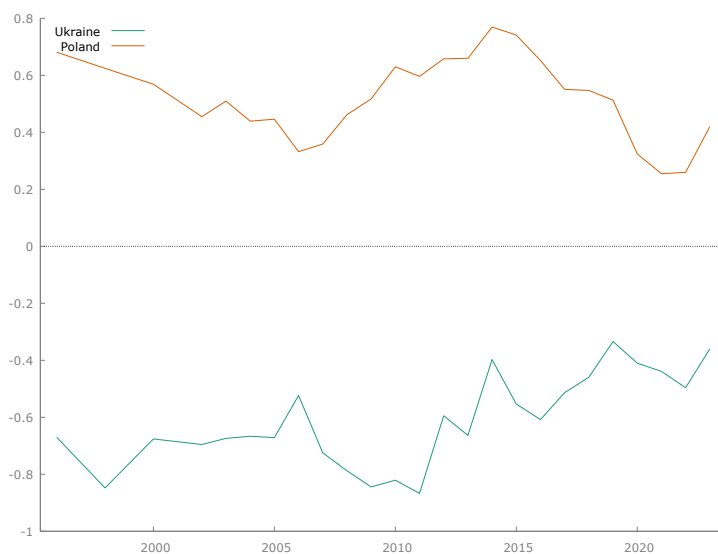


Data source: The World Bank

The graph illustrates global rankings for Ukraine and Poland in the Control of Corruption indicator over 1996–2022. Poland consistently remained in the upper quartile, averaging around the 70th–75th percentile, which suggests sustained positive perceptions of its institutional integrity. Ukraine, though starting from one of the lowest positions, has made incremental progress—from just 4.28% in 1998 to 29.25% in 2022. Despite the considerable gap, Ukraine’s upward trend signals a gradual improvement in its international standing on corruption control.

Let’s now turn to another key governance indicator — Government Effectiveness. This metric reflects the quality of public services, the professionalism and independence of the civil service, as well as the government’s capacity to formulate and implement sound policies. In the context of transition economies like Ukraine and Poland, the relevance of this indicator is hard to overstate. The more capable and politically independent the institutions, the more likely it is that reforms will be implemented effectively, and that public resources will be managed in a rational and transparent way.

Fig. 18 Government Effectiveness estimate index development in Ukraine and Poland during the period 1996–2023



The graph illustrates the evolution of the Government Effectiveness indicator in Ukraine and Poland from 1996 to 2023. Throughout the entire period, Poland consistently maintains positive values, peaking around 2015, followed by a slight decline. Ukraine, on the other hand, starts with low negative scores but shows steady improvement after 2014 — a trend likely linked to institutional reforms launched in the post-Euromaidan period. Although the average values

remain significantly different (around 0.50 for Poland and -0.60 for Ukraine), the gap between the two countries has gradually narrowed, suggesting some positive shifts in Ukraine’s administrative capacity over time.

Fig. 19 Government Effectiveness estimate index development in Ukraine during the period 1996-2023



Ukraine's Government Effectiveness index remained consistently low from 1996 to 2023, highlighting persistent weaknesses in public administration. Before 2005, scores were especially poor, reflecting political instability and weak institutions. A structural break appears in 2007, coinciding with growing political tensions and declining institutional trust, largely caused by the power struggle between President Yushchenko and Prime Minister Yanukovych. As Copsey (2008) notes, this conflict led to the dissolution of parliament and a sharp division between executive and legislative branches, effectively creating a dual power dynamic that stalled reforms and weakened state capacity. After 2014, the index shows a steady upward trend, aligned with reforms launched following the Revolution of Dignity. These included decentralization—drawing heavily from the Polish model (Malynovskyy, 2024)—enhanced fiscal control, and growing external pressure from the EU and IMF. While the score remains negative, its post-2014 trajectory reflects gradual progress in rebuilding Ukraine's governance structures.

As Copsey (2008) notes, this conflict led to the dissolution of parliament and a sharp division between executive and legislative branches, effectively creating a dual power dynamic that stalled reforms and weakened state capacity. After 2014, the index shows a steady upward trend, aligned with reforms launched following the Revolution of Dignity. These included decentralization—drawing heavily from the Polish model (Malynovskyy, 2024)—enhanced fiscal control, and growing external pressure from the EU and IMF. While the score remains negative, its post-2014 trajectory reflects gradual progress in rebuilding Ukraine's governance structures.

Fig. 20 Government Effectiveness estimate index development in Poland during the period 1996-2023

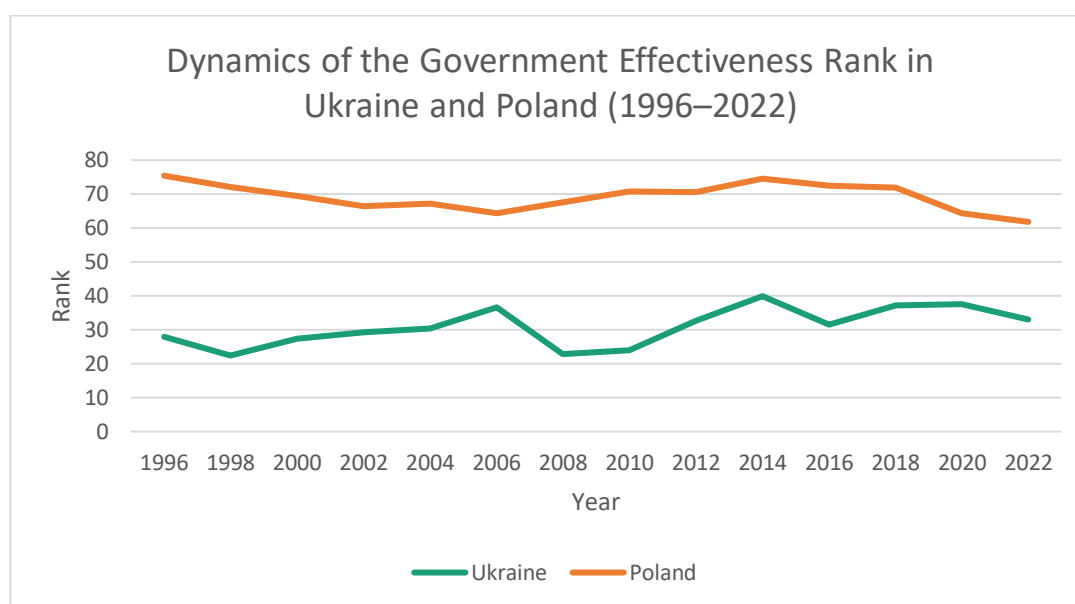


The graph traces the evolution of Poland's Government Effectiveness index from 1996 to 2023. In the initial years, the indicator showed a mild decline, likely reflecting institutional and economic pressures tied to the EU accession process. After Poland joined the EU in 2004, governance improved: institutional strength grew, transparency increased, and the index peaked around 2015. A gradual decline followed, potentially linked to internal

political shifts and EU concerns over judicial independence and rule-of-law issues. A key turning point came in 2010, which the model identifies as a structural break. That

year marked a brief surge in government effectiveness, followed by a slowdown. The 2010 Smolensk air disaster, which claimed President Lech Kaczyński and other senior officials, shocked the country but also revealed institutional resilience. As Kaczyński (2010) observes, the rapid response helped stabilize the political system, allowing Prime Minister Tusk’s cabinet to maintain continuity and push forward with modernization. Despite recent fluctuations, Poland continues to perform relatively well in governance when compared to other post-socialist states.

Fig.21 Dynamics of the Government Effectiveness Rank in Ukraine and Poland 1996–2022



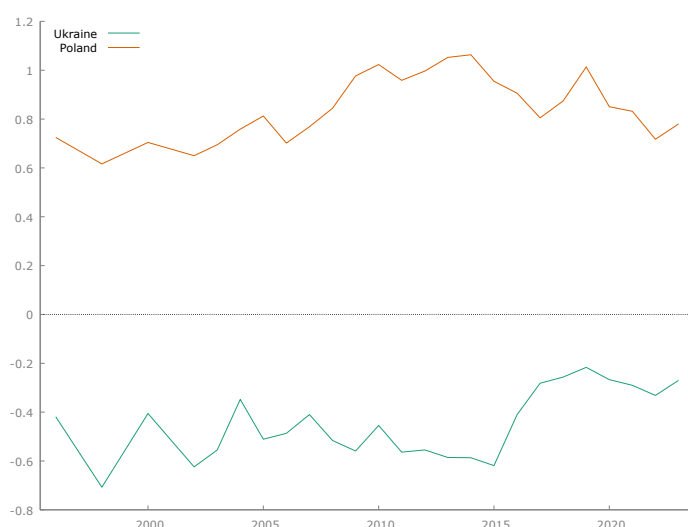
Data source: The World Bank

The graph shows the changes in Government Effectiveness Rank for Ukraine and Poland between 1996 and 2022, expressed in percentiles—higher values indicate stronger governance performance relative to other countries. Poland has consistently remained within the 60–75% range, with a notable improvement after joining the EU in 2004. Ukraine’s trajectory, on the other hand, has been more volatile. Following a sharp decline between 2008 and 2010, likely driven by political and economic turmoil, the country began to recover after 2014 amid institutional reforms and increased international pressure.

And finally — last but not least — the indicator that completes this part of the analysis is Regulatory Quality. This measure reflects the government’s ability to formulate and implement sound policies and regulations that enable private sector development.

It plays a crucial role in shaping the business environment: well-designed regulations can encourage investment, foster competition, and support sustainable growth. On the other hand, poor or excessive regulation often becomes a barrier to economic development, especially in countries undergoing structural transformation. Analyzing this indicator helps to better understand how institutional conditions have supported or hindered economic progress in Ukraine and Poland.

Fig. 22 Regulatory Quality estimate index development in Ukraine and Poland during the period 1996-2023



The graph shows the development of the Regulatory Quality index for Ukraine and Poland from 1996 to 2023. This indicator measures how effectively a government can design and apply regulations that support private sector growth. Poland consistently maintained positive scores throughout the period, peaking around 2015—largely due to institutional reforms and progress tied to EU integration. Ukraine, in contrast, stayed in the negative range until

2015. Since 2014, however, a steady upward trend is visible, likely reflecting reform initiatives and efforts to improve the business environment. Despite a notable difference in average scores—0.87 for Poland and -0.44 for Ukraine—the data suggests that the gap between the two has gradually narrowed.

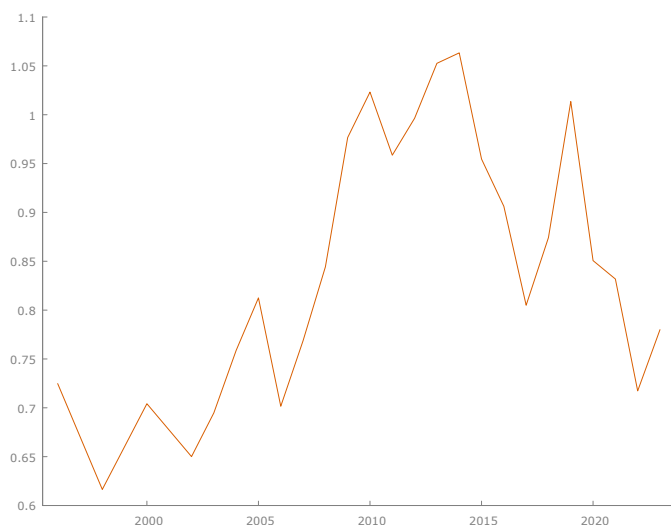
Fig. 23 Regulatory Quality estimate index development in Ukraine during the period 1996-2023



From 1996 to 2023, Ukraine's Regulatory Quality index remained negative, reflecting ongoing weaknesses in its regulatory system and administrative capacity. Prior to 2015, the indicator was unstable, likely due to frequent legislative shifts and political crises. A structural break in 2016 coincides with post-Revolution of Dignity reforms, including deregulation, business simplification, and legal alignment with EU standards.

While the average remains at -0.44, the model indicates a positive trend, suggesting gradual progress. A key step was the launch of the ProZorro e-procurement system to improve transparency and meet EU norms. As Balan and Bukovskiy (2021) note, ProZorro symbolized a shift toward accountable governance. This, along with repealing outdated regulations and simplifying permits, likely supported the rise in regulatory quality.

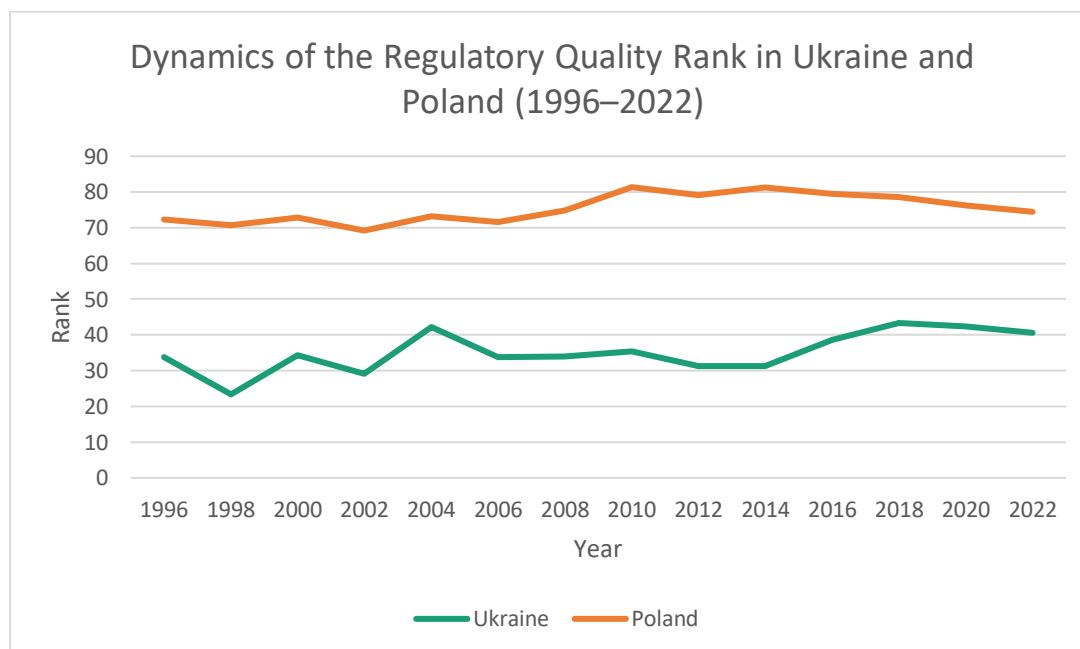
Fig. 24 Regulatory Quality estimate index development in Poland during the period 1996-2023



Poland's Regulatory Quality index remained positive from 1996 to 2023, reflecting a stable and improving regulatory system. A major shift occurred after 2004, as preparations for EU accession led to broad structural reforms aimed at aligning regulations with EU standards. A structural break in 2009 coincides with the global financial crisis, prompting new institutional and crisis-response efforts, including the *deregulacja gospodarki* strategy to ease administrative pressure on

businesses. As Égert and Goujard (2014) note, reforms at the time focused on simplifying firm registration, streamlining licensing, and cutting red tape. Poland also advanced digital transformation by expanding e-document workflows and enabling online business registration, which strengthened public administration and regulatory performance.

Fig.25 Dynamics of the Regulatory Quality Rank in Ukraine and Poland 1996–2022



Data source: The World Bank

The graph illustrates the dynamics of *Regulatory Quality* rankings for Ukraine and Poland. Throughout the entire observed period, Poland consistently holds positions in the upper tier of the ranking (typically within the 70th to 80th percentiles), reflecting a

stable regulatory environment and institutional maturity. Ukraine, in contrast, starts much lower but shows some improvement after 2014, likely driven by reform efforts and increased international pressure. Still, the gap between the two countries remains significant. Ukraine's rankings fluctuate more sharply, highlighting ongoing institutional instability and recurring political and economic crises. Since the rank is a relative indicator rather than a continuous one, the graph is intended for visual comparison only and is not used in regression analysis.

These were the most important institutional indicators within the scope of this research, as they directly reflect the quality of governance and its impact on long-term economic growth. In the following sections, additional indicators will be covered, though in a less detailed manner, to complement the broader picture of institutional development.

Fig.26 Dynamics of the Voice and Accountability Rank in Ukraine and Poland 1996–2022

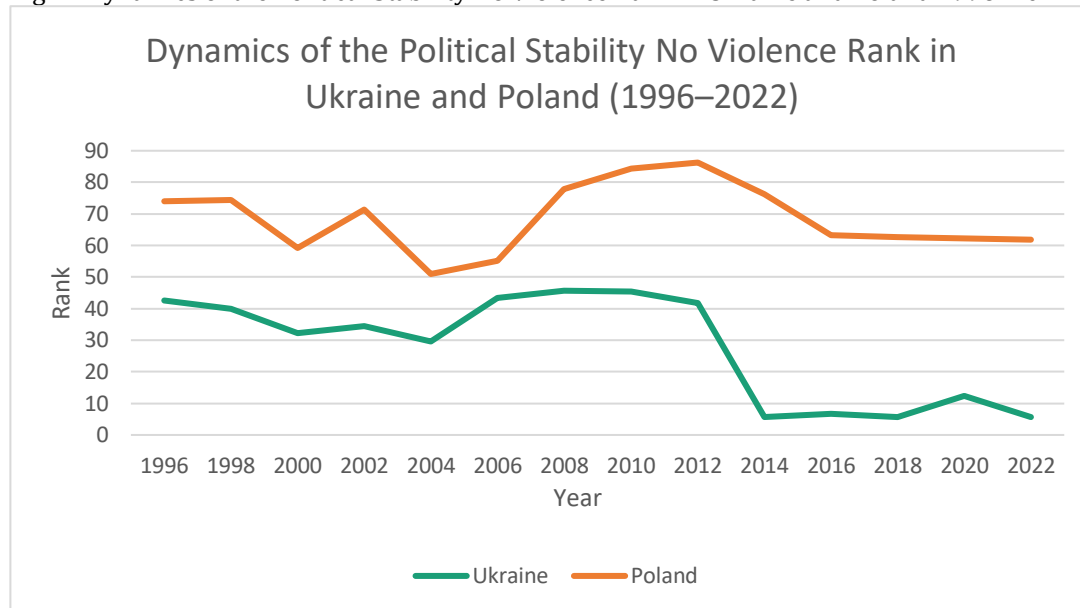


Data source: The World Bank

The Voice and Accountability indicator reflects how much citizens are able to participate in choosing their government, along with the levels of freedom of expression, media independence, and civil liberties. This metric plays a crucial role in assessing the quality of democratic institutions and the degree of public trust in the political system.

The chart shows that Poland has consistently held a strong position in global rankings for Voice and Accountability. Its rank remained high throughout the period, reflecting a stable democracy, active civil society, and strong protections for free speech. Ukraine, by contrast, displays a more uneven trajectory. After a decline in the early 2000s, the index improved following the 2005 Orange Revolution and again after 2014, amid rising political engagement and reform efforts. Nonetheless, Ukraine's ranking stays well below Poland's, underscoring ongoing difficulties in consolidating democratic institutions and ensuring transparency.

Fig.27 Dynamics of the Political Stability No Violence Rank in Ukraine and Poland 1996–2022

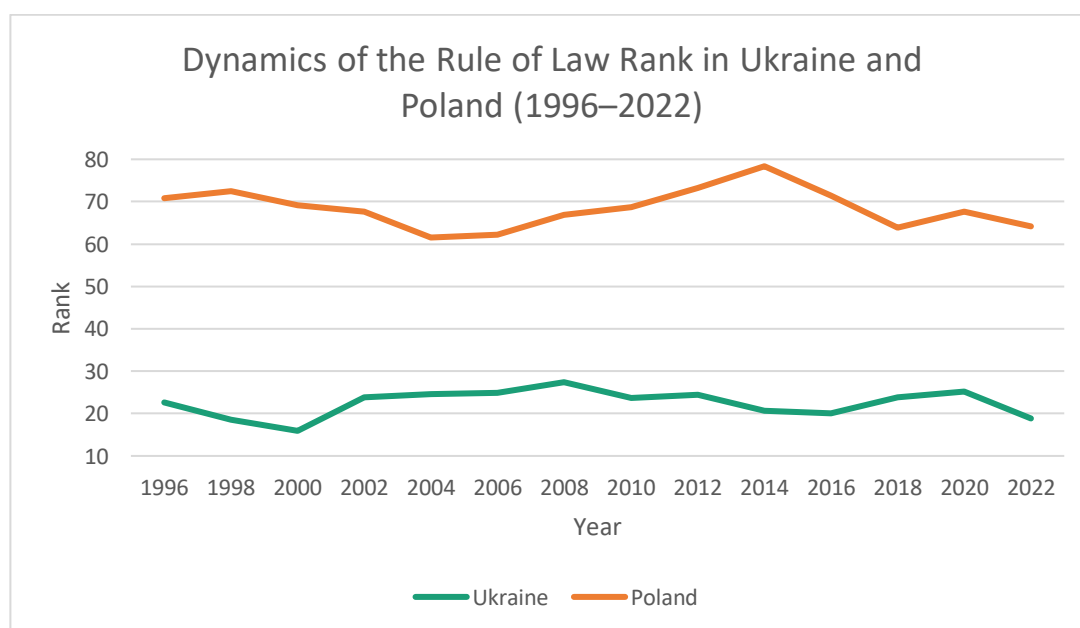


Data source: The World Bank

The Political Stability and Absence of Violence indicator reflects the risk of instability, conflict, terrorism, and unconstitutional power shifts. It is a key metric for evaluating institutional resilience, investment climate, and public confidence.

As the chart shows, the gap between Poland and Ukraine is especially wide in this area. Poland has maintained strong rankings, with only minor fluctuations in the early 2000s—indicating low political risk and stable institutions. Ukraine's path, by contrast, is far more fragile. A steep decline occurred after 2014, following the annexation of Crimea and conflict in the east. Although some recovery has been observed, persistent instability remains a major obstacle to Ukraine's long-term development.

Fig.28 Dynamics of the Rule of Law Rank in Ukraine and Poland 1996–2022



Data source: The World Bank

The Rule of Law indicator captures public confidence in legal institutions, including contract enforcement, property rights, policing, the judiciary, and protection from crime and violence. It is essential for sustaining long-term growth, legal certainty, and institutional trust.

According to the graph, Poland has consistently ranked relatively high, especially before 2015—a period shaped by active judicial reforms and EU-driven institutional changes. After 2015, a downward trend emerges, likely linked to legal reforms and disputes over judicial independence criticized by the EU. Ukraine, by contrast, consistently scores much lower, often ranking in the bottom third of countries. This reflects ongoing problems such as weak enforcement, limited judicial autonomy, and insecure property rights.

To bring together the key insights from the institutional analysis, the table below summarizes the main trends, structural breaks, and model results for the selected governance indicators examined in both countries.

Tab. 7 Summary regarding the models of time series for institutional quality and governance indicators

Indicator	Country	Observed period	Model	R ²	Break year	Trend shift	Conclusion
CPI (Corruption Perception Index)	Ukraine	2012-2024	OLS	0.92	-	Gradual improvement, no structural break	Ukraine's perception of corruption has been slowly improving over the past decade, likely due to external pressure and internal reforms, though the change has been incremental rather than sudden.
	Poland	2012-2024	OLS+ break	0.98	2016	Reversal from improvement to decline	While Poland initially improved, its CPI scores fell after 2016, reflecting rising concerns over political influence and weakening institutional independence.
Control of Corruption (WGI)	Ukraine	1996-2023	OLS+ break	0.8	2009	Initial decline, modest recovery post-2014	Institutional efforts and international support helped Ukraine begin reversing long-term corruption

							trends, though the system remains fragile.
	Poland	1997-2023	OLS+ break	0.78	2008	Fluctuations, weakening trend post-2015	Poland's strong anti-corruption image has faded slightly in recent years, partly due to internal political shifts.
Government Effectiveness (WGI)	Ukraine	1996-2023	OLS+ break	0.7	2007	Very low early levels, rising post-2014	Public administration in Ukraine saw major improvements after 2014, but historical weaknesses still affect overall effectiveness.
	Poland	1997-2023	OLS+ break	0.69	2010	Post-EU boost, slight backslide after 2015	Poland achieved gains in state capacity post-accession, yet political tension may have curbed further progress
Regulatory Quality (WGI)	Ukraine	1997-2023	OLS	0.57	2016 (implied)	Negative baseline, slow climb since reforms	Ukraine's regulatory framework is gradually catching up, though years of institutional lag are still visible.
	Poland	1996-2023	OLS+ break	0.82	2009	Strong initial trend, some slowdown post-2015	Poland built a solid base for regulatory performance, but recent years saw a plateau, hinting at institutional fatigue.

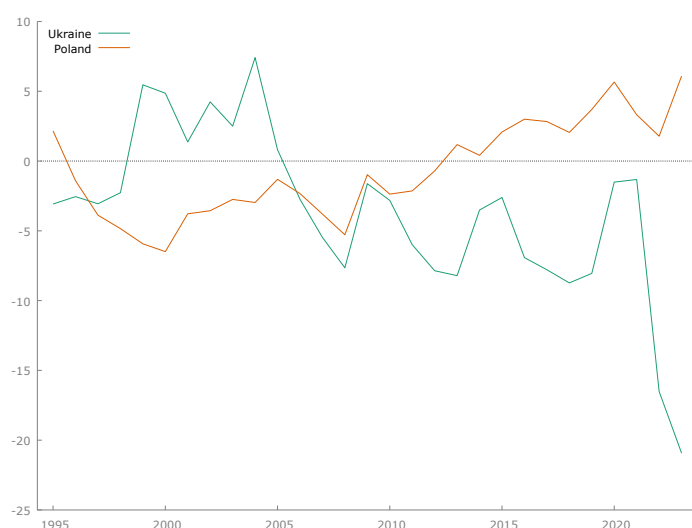
3.3 Trade and investments patterns

The next section focuses on two key indicators of external economic activity — trade balance and foreign direct investment (FDI) inflows in Poland and Ukraine. These indicators are essential for assessing how successfully each country integrates into the global economy, how open they are to international cooperation, and how effectively they utilize external resources to support domestic development. This is especially relevant

for transition economies, where foreign investments not only serve as a source of capital but also bring access to new technologies and modern management practices.

In this analysis, both indicators are expressed as a percentage of GDP. This relative format allows for a more accurate comparison between Poland and Ukraine, regardless of the size of their economies. It helps to understand how significant trade and investment flows are in relation to the overall economy, making the comparison clearer and more meaningful.

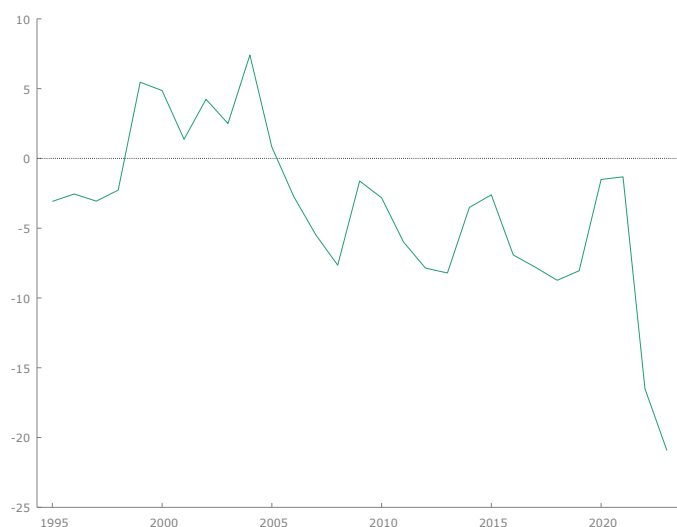
Fig. 29 Development of Trade Balance in Ukraine and Poland (as a % of the GDP) during the period 1995 – 2023



The chart shows trade balance trends as a percentage of GDP for Ukraine and Poland from 1995 to 2023. On average, Ukraine ran a deeper trade deficit (-3.61% of GDP), while Poland stayed closer to balance (-0.70%). A notable shift occurred in the 2010s, as Poland moved into surplus—driven by stronger exports, economic diversification, and deeper global integration. Ukraine, meanwhile, maintained a negative balance after 2014, with a sharp decline in 2022–2023 due

to war-related disruptions and falling exports. The data underscores structural differences in the external economic resilience of both countries.

Fig. 30 Development of Trade Balance in Ukraine (as a % of the GDP) during the period 1995 – 2023

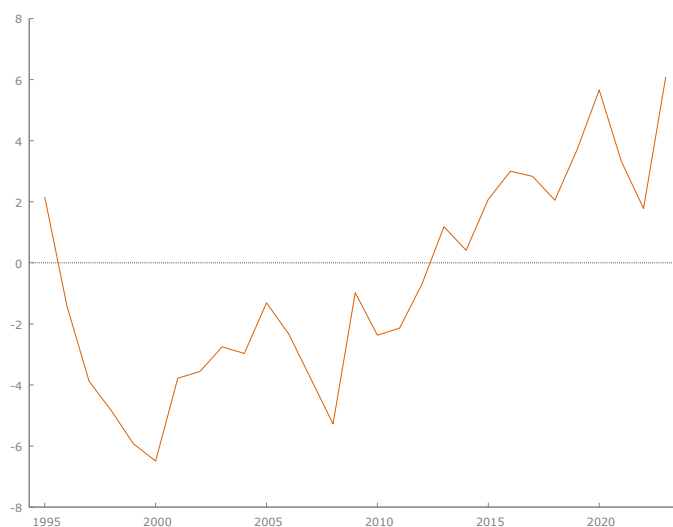


The chart indicates a steady decline in Ukraine's trade balance (% of GDP) since the mid-2000s. After relative stability in the 1990s, a clear deterioration began in 2005—a shift confirmed by a structural break in the model. Following the Orange Revolution, Ukraine moved from an export-oriented industrial model to one reliant on domestic consumption and foreign credit. As Mykhnenko and Swain (2010) explain, this shift triggered a surge in imports that outpaced domestic output.

The resulting imbalance increased dependence on foreign goods and worsened the trade deficit.

The most severe drop occurred in 2022 amid Russia's full-scale invasion, which devastated Ukraine's trade infrastructure and export capacity. Ports and transport routes were heavily damaged, while exports—especially metals and grain—collapsed. As Yevhen (2023) notes, the war also caused a spike in military and humanitarian imports, deepening the trade gap. Regression results confirm statistically significant breaks in both 2005 and 2022, marking key points in the worsening trend.

Fig. 31 Development of Trade Balance in Poland (as a % of the GDP) during the period 1995 – 2023



Since the mid-1990s, Poland's trade balance has steadily improved. Although the country initially ran sizable deficits, indicators turned consistently positive after the 2010s. This turnaround is linked to economic reforms, export growth, and deeper EU integration. A major boost followed EU accession in 2004, which expanded market access, spurred modernization, and enhanced competitiveness. Regression analysis identifies a

structural break in 1999, corresponding to external shocks like the Asian financial crisis and Russia's 1998 collapse—then a major export partner. As Égert and Goujard (2014) note, this period triggered a drop in trade with the East and a strategic shift toward importing capital goods from the EU. Expanding consumer credit also drove domestic demand, raising imports and widening the trade gap. According to Kowalski (2013), integration into global markets played a central role in Poland's economic transformation.

Though, the trade balance gives a general idea of how stable a country's external position is, it only scratches the surface. To truly understand what's behind the numbers, it's important to look at what countries export and who they export to. This kind of analysis becomes especially relevant in the case of Ukraine and Poland — two countries that started from similar positions in the 1990s but ended up taking very different paths.

For this comparison, three years were chosen: 1995, 2010, and 2023. Each of them captures a key phase — the starting point, a moment of transition, and the current situation. In 1995, both countries were only beginning their economic restructuring. By 2010, Poland was already part of the EU and positioning itself as an industrial player, while Ukraine, after the Orange Revolution, was just starting to look for a new direction. Finally, 2023 shows where the two countries have arrived after decades of reforms, political shifts, and economic realignments.

Ukraine and Poland's export trajectories since the 1990s tell the story of two very different development models. One evolved toward complexity and resilience; the other remained largely stuck in a raw material export loop. Despite starting from similar

conditions, their strategies for integration into the global economy diverged — and this is clearly reflected in their trade structures.

Ukraine's export story began in 1995 with a typical post-Soviet profile: hot-rolled iron, fertilizers, and semi-processed goods. The country still relied heavily on its Soviet-era industrial base, with minimal reform and chaotic privatization. Exports were mostly low in complexity and value-added. Even then, Russia was no longer the sole trade partner—Turkey, Italy, and Germany were already emerging markets—but raw, unprocessed goods remained dominant.

By 2010, a gradual shift toward agriculture became evident. Corn, wheat, and sunflower oil started to lead exports, while metals still played a role, though in decline. This transition pointed to some institutional progress, especially in agribusiness. Still, Ukraine's trade pattern remained inconsistent: Russia continued to hold a major share, and new ties with developing countries, though growing, were often unstable. The country was reorienting, but without a clear strategic pivot.

In 2023, the landscape has changed dramatically. Russia is no longer among Ukraine's top trading partners, replaced by Poland, Germany, China, and Turkey. This shift reflects not only the war but also new logistics and political alignment. Yet in terms of export composition, the core remains unchanged—agriculture now dominates even more. Apart from iron ore, industrial exports have nearly disappeared. Though sectors like IT are developing, they have yet to significantly reshape Ukraine's export structure, which still relies heavily on raw materials, now headed in different directions.

Poland took a different path. As early as 1995, it was already exporting processed goods—such as copper, furniture, clothing, and alcohol. The fact that most exports were directed toward EU markets highlights Poland's early alignment with European standards, even before its official EU accession.

By 2010, Poland was fully integrated into European production chains. Its export profile had shifted to include car parts, electronics, and appliances—signaling not just trade, but participation in Europe's broader industrial network. This transformation was supported by foreign investment, harmonized regulations, and consistent economic policy. Its core trading partners—Germany, France, and Czechia—remained stable, while the sophistication of exports steadily increased.

In 2023, Poland further advanced this trend. It now exports high-tech products like batteries, machinery, and vehicles—reflecting engineering capacity, innovation, and research. This shift stems from long-term reforms, sound fiscal strategy, and strategic use of EU funds. Poland no longer simply sells to the EU—it has become part of its industrial backbone.

So, when comparing these two countries, the contrast is hard to ignore. Poland has steadily climbed the ladder of value-added production and built itself into a manufacturing hub. Ukraine, despite some progress, still relies heavily on exports that require little or no processing. The trade flows have shifted, but the substance of what is being exported has yet to follow.

As a part of concluding the following table can be used.

Tab. 8 Comparison of Key Export Goods and Main Trade Partners: Ukraine and Poland (1995; 2010; 2023)

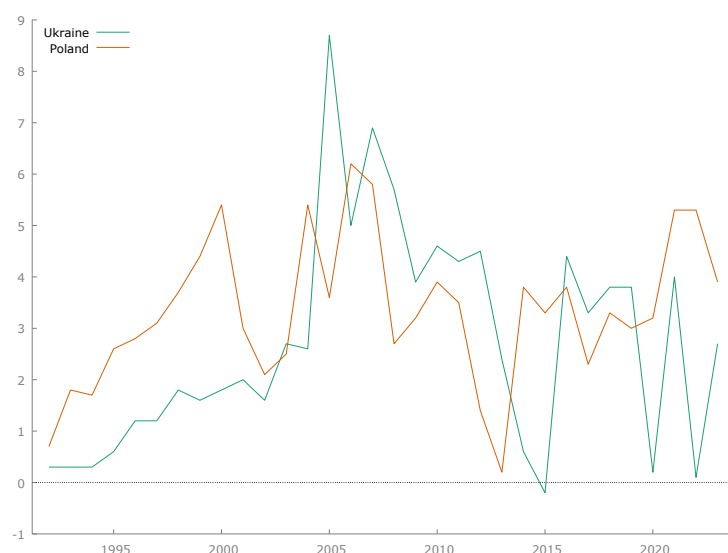
Year	Country	Key Export Goods	Main Trade Partners
1995	Ukraine	Hot-rolled iron, fertilizers, semi-products	Turkey, Italy, Germany, Russia

1995	Poland	Copper, furniture, clothing, alcohol	Mostly EU countries (Germany, etc.)
2010	Ukraine	Corn, wheat, sunflower oil, metals	Russia (major), Egypt, India, EU
2010	Poland	Car parts, electronics, appliances	Germany, France, Czech Republic
2023	Ukraine	Corn, sunflower oil, wheat, iron ore	Poland, China, Germany, Turkey
2023	Poland	Batteries, machinery, vehicles	EU countries (Germany as key partner)

Data source: The Observatory of Economic Complexity

While the trade balance highlights how countries perform in terms of exports and imports, it only tells part of the story. To fully understand a country's economic openness and integration into the global system, it is also crucial to examine foreign direct investment (FDI) inflows. FDI reflects long-term investor confidence, institutional quality, and the attractiveness of the local business environment. In transition economies like Ukraine and Poland, foreign investment plays a particularly important role, not just as a source of capital, but also as a vehicle for technological transfer, know-how, and structural transformation. Therefore, analyzing FDI inflows complements the trade balance assessment and offers a more comprehensive picture of external economic engagement.

Fig. 32 Development of FDI inflows in Ukraine and Poland (as a % of the GDP) during the period 1992 – 2023



The graph shows foreign direct investment (FDI) inflows as a share of GDP in Ukraine and Poland from 1992 to 2023. Poland demonstrates a steadier upward trajectory, especially after EU accession in 2004, which greatly enhanced its investment climate. Ukraine, by contrast, experienced significant volatility: inflows rose sharply after the Orange Revolution but dropped during the global financial crisis (2008–2009) and again amid the political

and military crises of 2014 and 2022. On average, Poland attracted 3.38% of GDP in FDI, while Ukraine received 2.71%, reflecting Poland's more stable and investor-friendly environment.

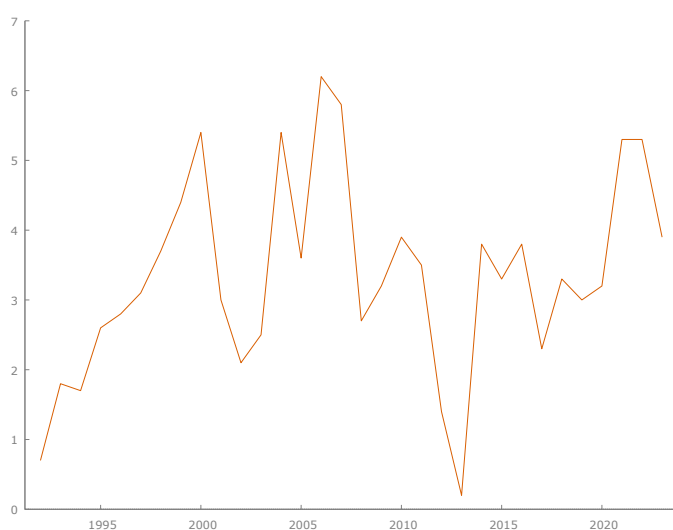
Fig. 33 Development of FDI inflows in Ukraine (as a % of the GDP) during the period 1992 – 2023



Ukraine's FDI inflows can be divided into several clear phases. A sharp rise in 2005 followed the Orange Revolution, when foreign investors—especially from the West—showed renewed interest. One key event was the re-privatization of Kryvorizhstal, the country's largest steel plant, sold to Mittal Steel for \$4.8 billion. As Paskhaver and Verkhovodova (2007) note, this deal marked a shift in Ukraine's privatization strategy and helped attract more foreign capital.

However, this positive trend didn't last long, likely due to the lack of deeper institutional reforms. In 2014, foreign investment dropped sharply after the annexation of Crimea, the war in Donbas, and the worsening economic crisis. As Iwański (2014) explains, the national currency collapsed, industry suffered, and investor confidence fell. FDI dropped by more than half compared to 2013. Overall, Ukraine's FDI trends seem to depend more on political events than on stable economic progress.

Fig. 34 Development of FDI inflows in Poland (as a % of the GDP) during the period 1992 – 2023



The chart shows that since the early 1990s, Poland steadily increased its FDI inflows, especially up to 2008—a period of strong investor interest and preparation for EU membership. The econometric model identifies a structural break in 2008, matching the global financial crisis. Although Poland handled the crisis relatively well, FDI still fell sharply. As Zimny (2012) notes, by 2010 investment was

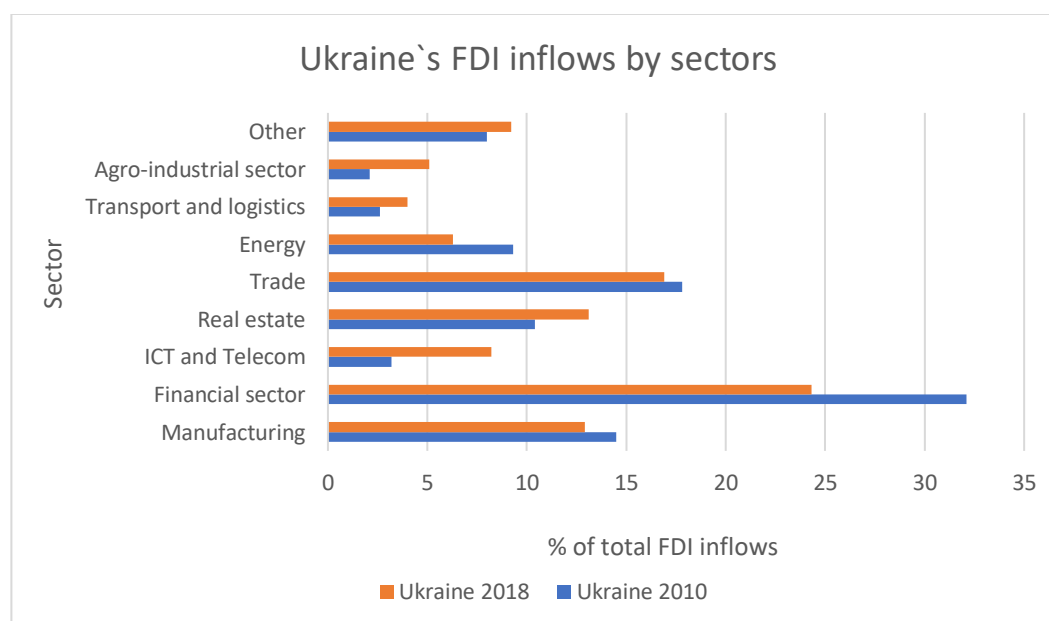
only 38% of its 2007 level, as multinational companies cut budgets due to rising global uncertainty. Despite the decline, Poland remained the most stable destination for investment in Central and Eastern Europe. After 2008, FDI kept growing, but the pace slowed. This change likely reflects a shift in global investment trends and the fact that Poland's economy had become more developed and predictable.

While tracking the overall dynamics of FDI inflows provides a general sense of how attractive a country is for foreign investors over time, it doesn't show *where exactly* this capital is going. To understand what type of economic growth these investments support — whether it's industry, services, or resource-based sectors — it's essential to look at the internal structure of FDI. The sectoral breakdown reveals not just the scale, but also the quality and direction of investment, helping to explain broader developmental differences between Ukraine and Poland.

Although 2010 and 2018 might not be the most conventional choices for analyzing FDI, they were the only two years for which consistent and comparable sectoral data was available for both Ukraine and Poland. At the same time, these years happen to capture important turning points in each country's economic path, making them particularly useful for understanding broader investment trends.

By 2010, Poland had already embraced its role within the EU, attracting stable investment in manufacturing and logistics. Ukraine, meanwhile, was in a relatively calm phase — post-Orange Revolution but still before the political upheavals of 2014. Fast forward to 2018, and both countries were adjusting to new realities: Ukraine was beginning to shift toward IT and services, while Poland had fully positioned itself within European industrial supply chains. Together, these two points in time offer a valuable snapshot of structural change — and highlight the different directions each country took.

Fig. 35 Ukraine's FDI inflows by sectors for 2010 and 2018



Data source: The World Bank

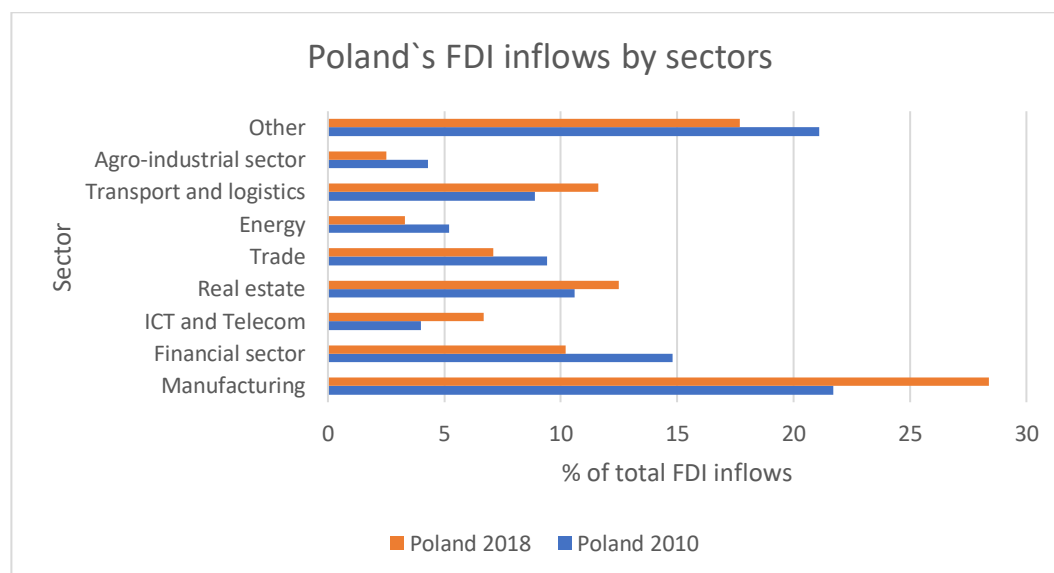
The chart clearly illustrates how the structure of foreign direct investment in Ukraine shifted over the span of eight years. Back in 2010, nearly one-third of all FDI was directed toward the financial sector — mostly banks, insurance companies, and similar services. That pattern made sense at the time: Ukraine was still relatively stable after the crisis, and financial services were seen as low-risk and potentially profitable by foreign investors. Trade and manufacturing were also among the top sectors receiving foreign capital.

By 2018, the picture had changed slightly. Although finance still held a large share, its dominance declined. At the same time, sectors like IT and telecommunications began to grow more rapidly — jumping from just 3.2% to 8.2%. Real estate and logistics also gained traction, suggesting a shift toward a more diversified economy and a growing interest in service-based industries.

However, manufacturing didn't strengthen — in fact, it lost ground compared to 2010. So even though there were some positive shifts, Ukraine still didn't manage to become a strong destination for investments in complex or high-tech production. Weak institutions, corruption, and general business risk continued to hold back long-term industrial FDI.

Overall, while some change is visible, it's hard to talk about a deep transformation. Ukraine remains largely reliant on a “traditional” FDI model — with capital flowing mainly into real estate, finance, and trade, rather than sectors that drive innovation or long-term growth.

Fig. 36 Poland's FDI inflows by sectors for 2010 and 2018



Data source: The World Bank

The chart clearly shows how, over the course of eight years, Poland strengthened its position as an attractive destination for industrial investment. Back in 2010, manufacturing already accounted for a solid 21.7% of total FDI, but by 2018 that share had grown to 28.4%. This increase points to growing interest from foreign investors in Poland's industrial base. And it's not surprising — the country had steadily integrated itself into European production chains and was becoming a major industrial hub in the region.

At the same time, the logistics and transport sector saw a noticeable increase in 2018, which makes sense given Poland's role as a key transit point within the EU. Real estate continued to attract stable investment, while IT and telecommunications slowly expanded their share — a reflection of ongoing digitalization and the development of tech clusters. Technological convergence accelerated Poland's integration into the European economy (Woźniak & Firszt, 2014).

The financial sector lost a bit of ground compared to 2010, but still remained important. Meanwhile, areas like energy and agriculture stayed on the sidelines of FDI — which, to be fair, mirrors broader trends in the European economy.

Overall, Poland presents a clear and well-structured model for attracting foreign investment. Thanks to the rising share of industrial, logistic, and technology-focused sectors, the country is gradually solidifying its role as an industrial center within the EU. Unlike Ukraine, where FDI remains concentrated in more traditional areas, Poland has made a visible shift over these eight years toward a more complex and value-added economic structure.

The following table summarizes the main developments in trade balance and FDI inflows, highlighting key shifts, structural changes, and overall patterns in how Ukraine and Poland have engaged with the global economy.

Tab. 9 Summary regarding the models of time series for trade and investment patterns indicators

Indicator	Country	Observed period	Model	R ²	Break year	Trend shift	Conclusion
Trade Balance (% of GDP)	Ukraine	1996–2023	OLS + breaks	0.85	2005,2022	Worsening trade gap after 2005, sharp fall in 2022	Ukraine's trade deficit deepened over time, with conflict-related disruptions in 2022 severely exacerbating an already negative trend.
	Poland	1995–2023	OLS + break	0.86	1999	From deficit to gradual surplus	Poland steadily transitioned from trade deficits to surpluses after late-90s reforms and EU integration, showing resilient export-driven growth.
FDI Inflows (% of GDP)	Ukraine	1992–2023	OLS + breaks	0.72	2005,2014	Spike post-2005, drop after 2014 and 2022	Ukraine's foreign investment has been highly unstable, swinging with political cycles and crises, rather than

							reflecting stable economic fundamentals.
	Poland	1992-2023	OLS + break	0.41	2008	Strong pre-crisis inflows, moderation afterward	Poland's investment growth cooled after the 2008 crisis, not due to domestic issues but as part of global FDI slowdowns amid maturing markets.

3.4 Economic Growth and Development Barriers: A Comparative Visualization

To provide a well-rounded comparison of the economic development of Ukraine and Poland, I decided to apply several analytical tools at once — a summary table, index charts, and scatter plots. This multi-layered approach allows for a deeper, more visually supported analysis and makes the conclusions clearer and more convincing. Since the focus of this study is on institutional aspects of development, it's important not just to list economic indicators, but to track their evolution and examine how they relate to economic growth.

Index charts are particularly useful for capturing long-term trends while accounting for initial disparities between the two countries. The summary table allows for a straightforward comparison of key figures, and scatter plots clearly illustrate the relationship between institutional quality and levels of economic development. In my view, this combination of tools is the most appropriate and informative for achieving the objectives of this research.

Let us begin with the Summary Comparison Table.

This table brings together key economic and institutional indicators for both countries, summarizing their average (and in some cases, median) values over different time periods. Such an approach offers a clear visual comparison, helping to highlight the most significant differences between Ukraine and Poland. Moreover, it lays the groundwork for further recommendations by drawing on Poland's more successful development experience.

Tab. 10 Summary comparison table for all analyzed indicators

Indicator (period)	Poland	Ukraine	Difference/Gap
GDP per capita in constant 2015 US\$ (1990-2023)	9888,21	2091,30	-7796,91
Inflation rate annual % (1993-2023)	7,65	12,80*	5,15

Unemployment rate % of total labor force (1991-2021)	11,17	7,86	-3,31
Corruption perception index (2012-2024)	58,15	30,31	-27,85
Government effectiveness WGI estimate(2002-2023)	0,50	-0,60	-1,10
Regulatory quality WGI estimate (2002-2023)	0,87	-0,44	-1,31
Control of corruption WGI estimate (2002-2023)	0,51	-0,92	-1,43
Trade balance as % of GDP (1995-2023)	-0,70	-3,61	-2,91
FDI inflows as % of GDP (1992-2023)	3,38	2,71	-0,67

*Unlike the other indicators, inflation for Ukraine was calculated using the median rather than the mean. This adjustment was necessary due to the period of hyperinflation in the early 1990s, when extremely high values would have significantly distorted the average. For reference, if the mean had been used, the result would have been 195.97

The summary comparison table clearly highlights the gap between Poland and Ukraine, both in terms of economic and institutional indicators. One of the most striking differences is GDP per capita — in Poland, it is nearly five times higher than in Ukraine. This discrepancy reflects not only differences in economic growth but also in institutional quality, investment climate, and governance effectiveness.

Even basic macroeconomic indicators, such as inflation, reveal a notable contrast. Ukraine has faced persistently higher inflation rates over the decades — even after excluding extreme values from the 1990s — pointing to a less stable monetary framework. While Ukraine reports lower average unemployment, Poland's higher historical figures should be viewed through the lens of the difficult transition period in the 1990s, which was followed by successful labor market stabilization.

What stands out most are the institutional indicators and corruption perception scores. Ukraine remains in negative territory across key governance dimensions, while Poland consistently shows strong, positive values. This reflects Poland's steady progress with institutional reforms, whereas Ukraine continues to struggle with instability and corruption risks. These differences are also reflected in trade balance and foreign direct investment inflows — both of which are more favorable in Poland, largely thanks to EU integration and a more supportive business environment.

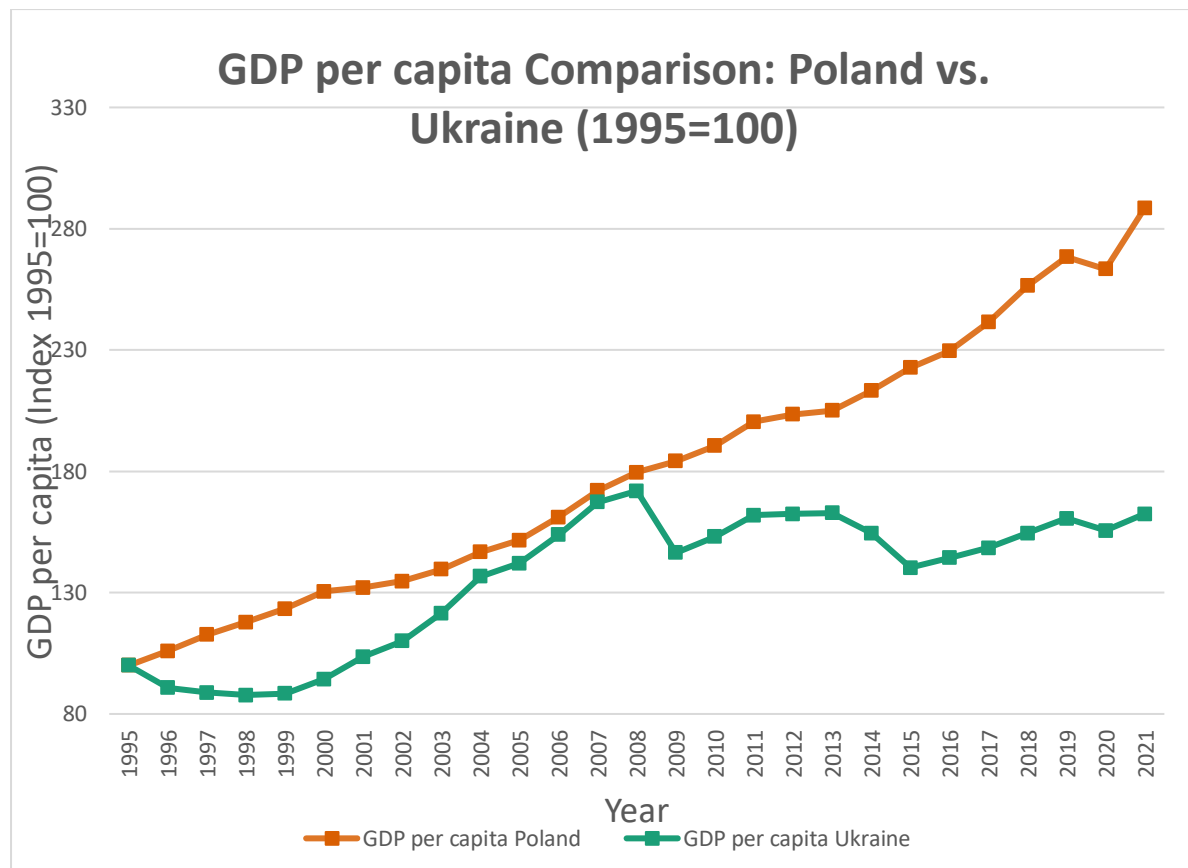
Overall, the table underscores a crucial conclusion: institutional quality is a fundamental factor behind the divergence in economic development trajectories between the two countries.

The next stage of the analysis focuses on Index Charts. This tool is particularly effective in comparative economic studies, as it allows for a clear visualization of how indicators evolve over time relative to a base year. By normalizing the data, index charts help highlight trends even when absolute values differ significantly across countries. In the context of my research — which aims to understand the roots of Poland's success and

identify key barriers to Ukraine's development — this approach proves especially valuable.

To explore these trends, I selected four core indicators: GDP per capita, FDI inflows, Government Effectiveness, and Control of Corruption. These indicators are directly linked to the main research questions and provide a solid basis for comparing the economic and institutional trajectories of both countries.

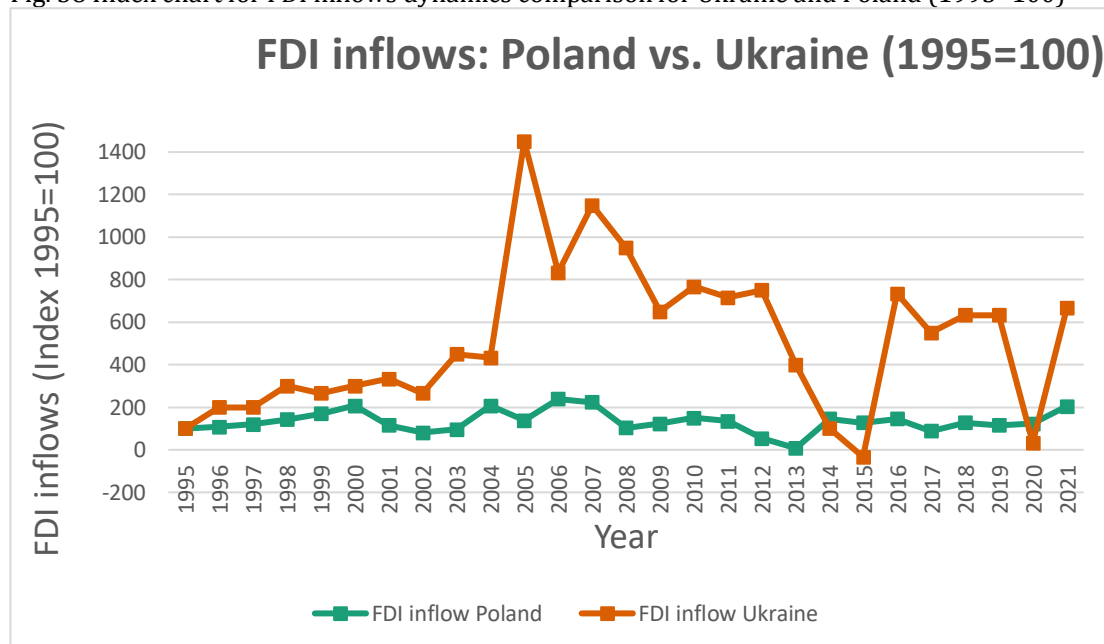
Fig. 37 Index chart for GDP per capita dynamics comparison for Ukraine and Poland (1995=100)



Data source: The World Bank

The index chart of GDP per capita clearly shows a growing divergence between Ukraine and Poland starting in the early 2000s. Poland's sharp upward shift after 2004 aligns with its EU accession, which triggered major reforms, a surge in investment, and broader international integration. Meanwhile, Ukraine's trajectory was repeatedly disrupted by political instability, economic crises, and ongoing conflict, preventing it from establishing steady long-term growth.

Fig. 38 Index chart for FDI inflows dynamics comparison for Ukraine and Poland (1995=100)



Data source: The World Bank

The chart illustrating FDI inflows once again highlights the contrast in economic stability between the two countries. In Ukraine, foreign direct investment fluctuates sharply—periodic surges are followed by steep declines, largely due to internal instability and weak investor protection. In contrast, Poland shows a more stable upward trend, particularly after joining the EU. This improvement can be attributed to transparent regulatory policies and a more predictable investment environment.

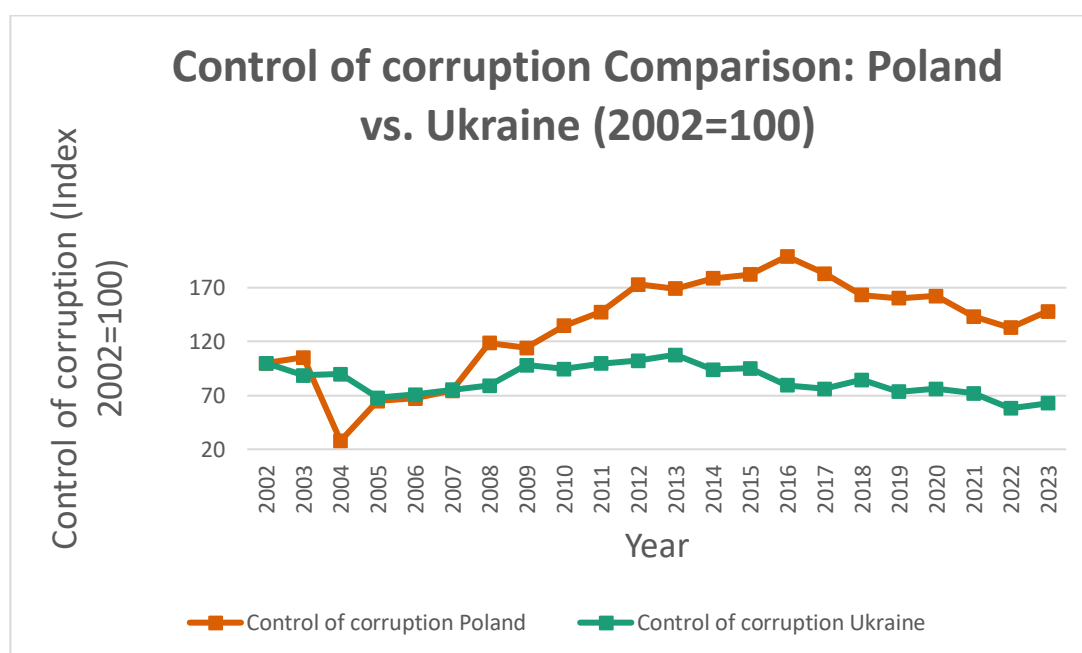
Fig. 39 Index chart for Government Effectiveness indicator dynamics comparison for Ukraine and Poland (2002=100)



Data source: The World Bank

The Government Effectiveness chart shows a steady improvement in the quality of governance in Poland up until the mid-2010s, followed by a moderate decline likely linked to domestic political changes. However, the overall level remains consistently high. In Ukraine's case, the indicator lacks a clear upward trend, pointing to institutional weakness and a lack of consistent reform efforts—particularly during periods of political instability.

Fig. 40 Index chart for Control of Corruption indicator dynamics comparison for Ukraine and Poland (2002=100)



Data source: The World Bank

The Control of Corruption chart clearly highlights the institutional gap between Poland and Ukraine. Poland has made notable progress, especially after joining the EU, thanks to systematic anti-corruption efforts and the strengthening of independent institutions. Ukraine, on the other hand, has remained in the low-performance zone throughout the entire period. Even after the post-2014 reforms, public trust in anti-corruption institutions remains weak, indicating the limited depth and consistency of those changes.

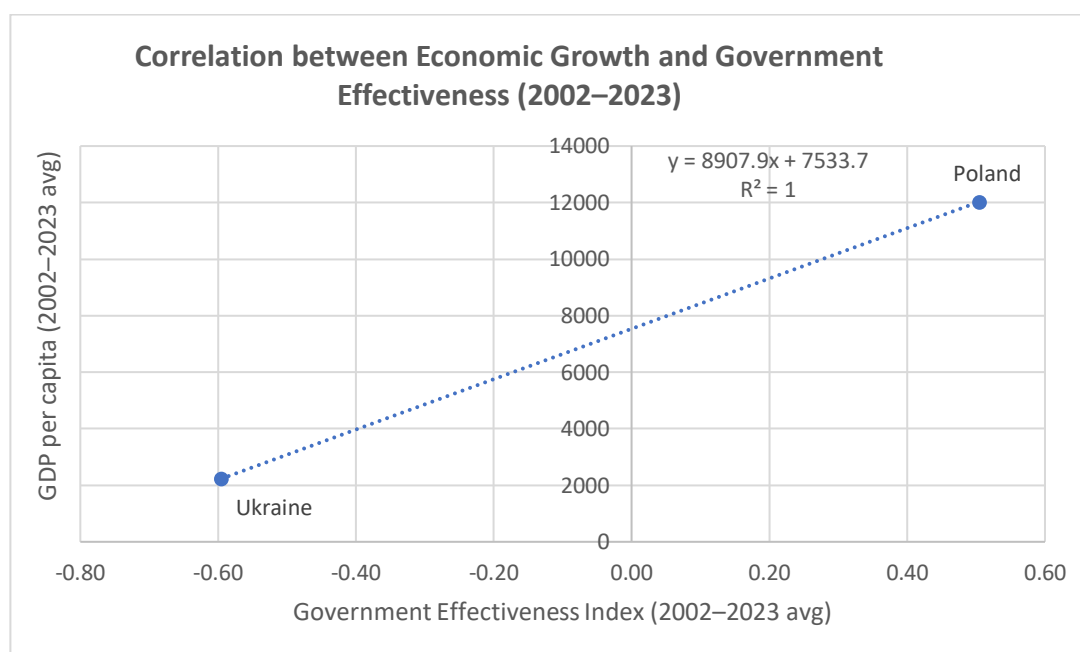
Overall, the index-based analysis leads to a key conclusion: the quality of institutions, transparency in governance, and predictability of economic policy have played a decisive role in Poland's success. For Ukraine, institutional fragility and poor governance continue to act as major roadblocks to sustainable growth. The indexed approach helped visualize not just the differences, but how those institutional factors directly shape long-term development paths—providing a solid base for future reform strategies.

Let us now turn to the Scatter Plot analysis. This part of the study plays a crucial role, as it allows for a visual understanding of the relationship between government effectiveness and the level of economic development. It helps explain why Poland has achieved steady growth, while Ukraine continues to face serious economic challenges.

To build the first scatter plot, two key indicators were selected: the average GDP per capita and the average Government Effectiveness score over the period 2002–2023. This timeframe captures all major events — including Poland's accession to the EU, the global financial crisis, and the political upheavals in Ukraine — making the analysis more comprehensive and balanced.

The choice of these indicators is deliberate, as they reflect the link between institutional quality and economic performance. Poland's consistently high Government Effectiveness scores suggest that the state has been able to use its resources wisely and create an environment conducive to growth. In contrast, weak institutions in Ukraine have remained one of the main obstacles to long-term economic progress.

Fig. 41 Scatter plot for Correlation between Economic Growth and Government Effectiveness for Ukraine and Poland (2002–2023)



Data source: The World Bank

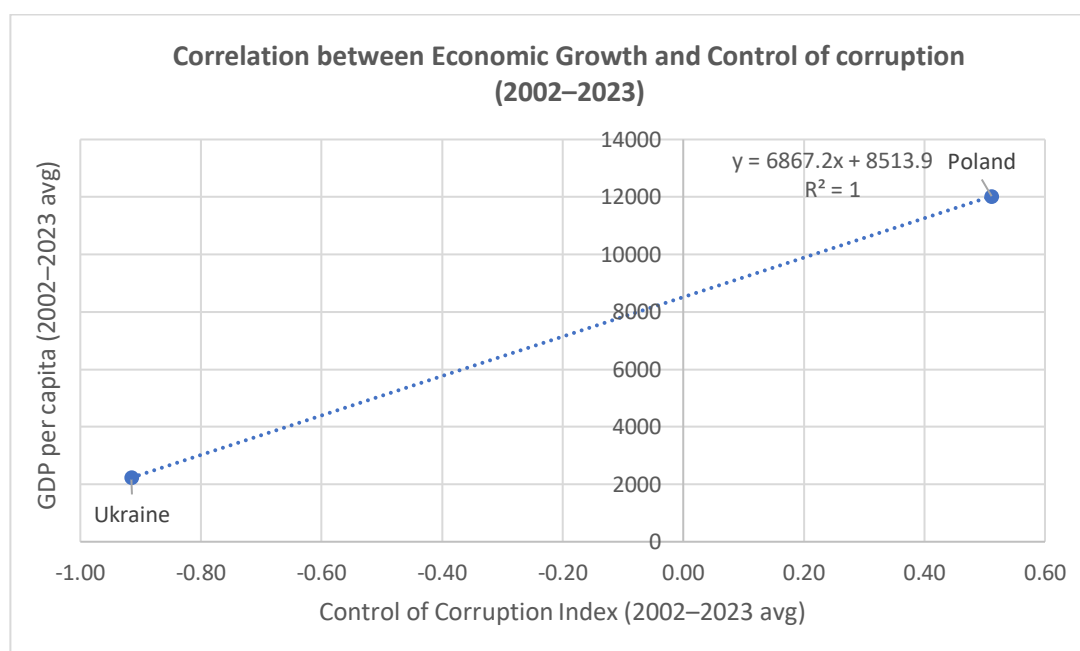
The scatter plot clearly illustrates a strong positive relationship between the quality of government and the level of economic development. Poland, with a significantly higher Government Effectiveness score (around 0.5), also demonstrates a much higher average GDP per capita — over \$12,000. Ukraine, by contrast, is located in the lower-left corner of the graph, with a low effectiveness score (around -0.6) and a correspondingly low GDP per capita of just over \$2,000. This stark contrast highlights how deeply institutional quality can influence economic outcomes.

The linear trend line built on these two observations shows an almost perfect correlation ($R^2 = 1$), which strongly suggests that differences in government effectiveness are not just a background factor — they may be central to understanding the development gap between the two countries. However, it's essential to point out that this correlation is based on only two cases — Ukraine and Poland — so it reflects a very specific comparison rather than a universal rule. Still, it provides compelling support for

one of the core arguments of this research: strong institutions and effective governance are critical drivers of long-term economic growth.

In Ukraine's case, improving the quality of governance — from public administration to regulatory efficiency — could unlock much of the country's economic potential. Without such progress, even well-designed macroeconomic strategies may fall short due to implementation issues and lack of institutional trust. Meanwhile, Poland's trajectory serves as a clear example of how investment in institutional development can translate into real and sustained economic gains.

Fig. 42 Scatter plot for Correlation between Economic Growth and Control of Corruption for Ukraine and Poland (2002–2023)



Data source: The World Bank

The graph illustrates the relationship between the average level of corruption control and economic development in Ukraine and Poland from 2002 to 2023. Poland is positioned in the upper-right corner of the scatter plot — indicating both a high level of corruption control and a high average GDP per capita. Ukraine, on the other hand, is located in the lower-left corner, reflecting weak anti-corruption institutions and lower income levels.

The trend line highlights a strong positive correlation between the two indicators. While the R^2 value of 1 results from comparing only two observations, it visually reinforces how drastically different these countries are in terms of institutional quality. According to the regression equation, a one-unit improvement in corruption control is associated with a potential increase of nearly \$6,867 in GDP per capita.

This graph reinforces one of the study's central conclusions: corruption seriously hampers economic development, while efforts to reduce it create a foundation for sustainable growth. Poland's experience shows how institutional reforms and enhanced transparency can generate long-term economic benefits.

Together, both scatter plots clearly emphasize the role of institutional factors in shaping development outcomes. The gap between Ukraine and Poland becomes especially apparent when we look beyond just GDP figures and focus on what drives them — governance quality and corruption control. Over the past two decades, Poland has managed to build strong and effective institutions, which directly supported its economic progress. Ukraine, although making some headway, still struggles with systemic institutional challenges.

This kind of visual analysis strengthens the broader argument of the thesis: economic growth is not only about money or GDP — it's also about competent, accountable governance and public trust. For Ukraine, this means that real progress will require not just economic measures, but deep institutional reforms in public administration and anti-corruption efforts. Without these, the gap with Poland is unlikely to close.

4 Recommendations

Throughout this thesis, I have tried to show that the success of economic transformation is not just about growth rates or export numbers. What matters most is the strength of a country's institutional foundation. As Douglas North (1990) emphasized, sustainable development is only possible when the “rules of the game”—both formal and informal institutions—reduce uncertainty and create incentives for productive economic activity. At this point, Ukraine remains caught in a cycle of weak institutions, and that's why any reform strategy must begin with fixing what lies at the base and then building upward.

The first and arguably most important step is to launch a real institutional reform aimed at reducing corruption and rebuilding trust in public institutions. This is more than just a formal goal. According to Minakov & Rojansky (2015), political instability and oligarchic influence have been among the biggest barriers to stable economic growth in Ukraine. Strengthening anti-corruption efforts, improving the transparency of public procurement, and expanding digital services (such as the “Diia” platform) aren't simply trendy policies—they're essential prerequisites for everything else to work.

Once this foundation is in place, the next logical move is to reform the judicial system and ensure stronger property rights. As Gylfason et al. (2022) point out, Ukraine's weak legal institutions are one of the main reasons foreign investors stay away. Without confidence that their capital will be protected from arbitrary interference or corporate raiding, sustainable FDI growth is out of reach. The solutions here are clear: transparent and competitive selection of judges, real independence for anti-corruption institutions, and—perhaps most importantly—political will demonstrated through action, not just words.

The next priority is to restart the privatization process and restructure state-owned enterprises, with a focus on attracting strategic investors and promoting competition. Adarov et al. (2015) highlight that one of Poland's key advantages was its focus on building a competitive private sector—not just redistributing assets among powerful actors. Ukraine should revisit past privatization decisions and establish a process that is transparent, internationally monitored, and oriented toward long-term value.

Another vital area is reforming the education system and addressing the growing gap in workforce qualifications—especially in engineering and technical fields. As both this research and Zvi Lerman (1999) emphasize, the outflow of human capital is putting Ukraine's chances of building an innovation-based economy at serious risk. More investment in universities, the development of dual education systems, and stronger links between business and academia could yield tangible results even in the medium term.

At the same time, infrastructure and logistics must be upgraded if Ukraine wants to fully integrate into global value chains. Here, Poland's experience is particularly relevant: becoming a logistics hub for the EU didn't happen overnight. Expanding Ukraine's port, rail, and customs infrastructure—especially at EU borders—could significantly enhance export capacity, not just for agriculture but for industry too.

Lastly, the government should not overlook the IT and creative sectors, which are among the most dynamic parts of the economy and could help reduce Ukraine's dependency on raw material exports. Yet, as the empirical data shows, their share in total FDI remains too small. This calls for targeted support: tax incentives, tech incubators, export assistance, and better cooperation with private investment funds that specialize in scaling up innovative companies.

Taken together, these actions form a step-by-step strategy: starting with institutional cleanup and trust-building → moving to investor protection and competitive markets → followed by investments in human capital → then strengthening physical integration into European supply chains → and finally, making the leap toward a digital and innovation-driven economy. Each part of this chain is supported by academic insights and real-world evidence. If implemented consistently, this path could help Ukraine finally break free from the “raw material trap” and set the country on course toward sustainable, inclusive growth.

5 Discussion and conclusion

The main objective of this thesis was to understand why Poland and Ukraine, starting their economic transitions from relatively similar conditions, ended up on such different development paths. Using a combination of quantitative tools—ranging from econometric modeling in Gretl to export and FDI structure analysis—alongside a review of relevant academic literature, the study sought to answer two main questions: (1) what were the main challenges Poland and Ukraine faced during their transitions, and how did these differ; and (2) what factors contributed to Poland's success, and how could those lessons help shape recommendations for Ukraine's development?

The findings suggest that the most decisive factors were not simply macroeconomic policies or access to international support, but rather institutional stability and the consistency of reform strategies. From the early 1990s, Poland focused on building a predictable legal and institutional framework that protected property rights and enabled market-oriented behavior. These foundations built the way for broader transformation, sustained investment inflows, and successful integration into European industrial networks.

Ukraine, on the other hand, followed a different trajectory. Despite certain breakthroughs—especially in agriculture and IT—the overall institutional environment remained fragile and inconsistent. The analysis of structural breaks, export composition, and FDI allocation made this pattern clear: Ukraine still relies heavily on raw material exports, while investment flows largely target low-value sectors such as finance, real estate, and retail.

Part of this situation can be attributed to external challenges: the absence of the EU integration incentives, geopolitical tensions, and the impact of war and instability. However, the internal factors may have played an even greater role. Weak governance, corruption, and a lack of judicial independence have all contributed to an environment where long-term economic planning is difficult. As Douglas North (1990) argues, institutions aren't just formal rules—they shape expectations and behavior. Without a solid institutional core, it becomes nearly impossible to sustain structural change.

One of the more telling observations from this study was the evolution of Ukraine's trade orientation. While the country shifted its trade geography away from Russia and toward the EU and Asia, the nature of its exports didn't change much. The same applies to FDI: the partners may have changed, but the targeted sectors remained largely the same. This suggests that international reorientation alone doesn't lead to meaningful transformation unless it's supported by internal reforms.

It's important to acknowledge the limitations of this study. For some indicators—especially institutional metrics—the time series were relatively short, limiting the scope of econometric modeling. Structural breaks could, in some cases, be influenced by external shocks that aren't easily quantifiable. Furthermore, this research couldn't fully address issues like migration, the role of human capital, or the informal economy due to data constraints. These gaps offer important avenues for future research.

At the same time, one of the key strengths of this thesis was its attempt to connect empirical analysis with historical and political context. Rather than treating numbers in isolation, the goal was to interpret them in light of the broader reform narrative. This is particularly relevant when analyzing institutions, which often fall outside the scope of conventional economic models. Theoretical perspectives from North, along with insights

from Douarin, Mickiewicz, and Adarov, helped frame the findings within a broader discussion about reform sequencing and institutional depth.

From a practical point of view, this research not only highlights where Ukraine and Poland diverged, but also outlines a potential roadmap for Ukraine. The recommendations proposed in the final chapter reflect a phased approach: starting with institutional clean-up and property rights enforcement, followed by industrial development and support for innovation-driven sectors. Each step is grounded not only in the data, but also in examples of what worked in Poland's case.

Ultimately, this study reinforces a crucial insight: economic development is not just about GDP growth or export statistics. It is about institutions, trust, reform coordination, and long-term vision. Poland serves as a case study in how consistent reform can deliver real results, even under limited resources. Ukraine, in contrast, remains a country with enormous potential—yet unless that potential is supported by credible and functioning institutions, it is unlikely to be fully realized.

This thesis does not offer a magic formula. But it does point to a direction. If Ukraine wants to break free from its institutional trap and come closer to the standards of sustainable European growth, it must follow a long, yet necessary path: from institutional reform to the construction of a new economic architecture that can support the development of high-value sectors. The sooner this process begins, the higher the chances that Ukraine's future will not be defined by divergence—but by convergence.

6 References

ADAROV, A., ASTROV, V., HAVLIK, P., HUNYA, G., LANDESMANN, M., PODKAMINER, L. (2015). *How to Stabilize the Economy of Ukraine*. Vienna: Vienna Institute for International Economic Studies. ISBN 978-3-902999-01-1.

ARI, A., PULA, G. (2021). *Assessing the Macroeconomic Impact of Structural Reforms in Ukraine*. IMF Working Papers. [online]. DOI: 10.5089/9781513582580.001. [cited 2025-03-25].

BALAN, A., BUKOVSKIY, D. (2021). "Prozorro" Electronic System as a Revolution in the Public Procurement System in Ukraine. *Economics: Time Realities*.

BALCER, A. (2016). *The Polish "Good Change" – What Does It Mean for Relations with Germany and the Rest of Europe?*

BALCEROWICZ, L. (2022). *Poland: Economic Reforms Under Extraordinary and Normal Politics—An Insider's View*. Budapest: Central European University Press. ISBN 978-9633864528.

BELLAS, C. J., BOCHNIARZ, Z., JERMAKOWICZ, W. W., MELLER, M., TOFT, D. (1994). *Foreign Privatization in Poland*. Warsaw: Center for Privatization Studies.

BERENSON, M. (2018). *Reinterpreting History to Recreate the State: The Transformation of the Polish, Russian and Ukrainian State Bureaucracies in the 1990s and Today*. Cambridge: Cambridge University Press.

BOYARCHUK, O. (2019). *The Economic Transition of Post-Soviet Countries: Comparative Analysis*. Kyiv: Institute for Economic Research.

BRESLAUER, G. W. (2021). *The Rise and Demise of World Communism*. Oxford: Oxford University Press. ISBN 978-0197573398.

BUKOWSKI, M., LEWANDOWSKI, P., MAGDA, I. (2007). *Employment in Poland 2006: Productivity for Jobs*. Warsaw: Polish Ministry of Labour and Social Policy.

BURDIAC, V. (2014). Polish Factor in Formation and Elaboration of the European Union's Policy in Regard to Ukraine. *Studium Europy Środkowej i Wschodniej*.

COPSEY, N. (2008). The Ukrainian Parliamentary Elections of 2007. *Journal of Communist Studies and Transition Politics*, 24(3), pp. 297–309.

CZYŻEWSKI, B., MATUSZCZAK, A. (2020). Changes in the Structure of Budget Expenditure on the Agricultural Sector in Poland in the Passing Financial Perspective 2014–2020. *Zagadnienia Ekonomiki Rolnej*, 4(365), pp. 3–24.

DOLBNIEVA, D. V. (2019). Ways of Transformation of the System of Combating Economic Crimes in Ukraine Considering EU Experience. *Law and Society Journal*.

DOLIŃSKA, A. (2019). Socio-Economic Costs of Systemic Transformation in Ukraine in the Lens of the Biographical Experiences of Ukrainian Female Migrants to Poland. *Qualitative Sociology Review*.

DOUARIN, E., MICKIEWICZ, T. (2017). *Economics of Institutional Change: Central and Eastern Europe Revisited*. London: Palgrave Macmillan

ÉGERT, B., GOUJARD, A. (2014). Strengthening Competition in Poland. *OECD Economics Department Working Papers*. [online]. [cited 2025-04-12].

FIDRMUC, J. (2001). Democracy in Transition Economies: Grease or Sand in the Wheels of Growth? *European Economic Review*, 45(8), pp. 1377–1400.

FURGACZ, P. (2020). Polish Policy Toward Ukraine After the Weighty Geopolitical Changes of the Dignity Revolution.

GEVORKYAN, A. V. (2018). *Transition Economies: Transformation, Development, and Society in Eastern Europe and the Former Soviet Union*. London: Routledge.

GYLFASON, T., HOCHREITER, E., KOWALSKI, T. (2022). Different Choices, Divergent Paths: Poland and Ukraine. *Comparative Economic Studies*, 64(3), pp. 456–475.

HAVRYLYSHYN, O., IVASCHENKO, I., MCGETTIGAN, D. (2002). *Economic Growth in Ukraine: What if Reforms and Economic Recovery Had Started Earlier?* Washington, D.C.: IMF Institute.

HÜBNER, D. (2004). Impact of the Membership in the European Union on Economic Growth in Poland. *European Integration Studies*.

IWAŃSKI, T. (2014). Ukrainian Economy Overshadowed by War. *OSW Commentary*, No. 148.

JACKSON, J. E., KLICH, J., POZNAŃSKA, K. (2002). Democratic Institutions and Economic Reform: The Polish Case. *British Journal of Political Science*.

KACZYŃSKI, P. (2010). *Pain in Poland*. CEPS Commentaries, 13 April.

KASIANOV, G. (2023). In Search of Lost Time? Decommunization in Ukraine, 2014–2020. *Journal of Modern Ukrainian Studies*.

KAMPO, V. M. (2018). Was Constitutional Democracy Necessary for the Establishment of a Socially Oriented Market Economy in Ukraine in 1994–2004?

KOSHULKO, O., KOSHULKO, V. (2016). The Role of Ukrainians in the Economic Growth of Poland. *Development Economics: Macroeconomic Issues in Developing Economies eJournal*.

KOWALSKI, T. (2013). Globalization and Transformation in Central European Countries: The Case of Poland.

KRAVCHUK, R. (1998). Budget Deficits, Hyperinflation, and Stabilization in Ukraine, 1991–96. In: FISCHER, S., SAHAY, R. (eds.). *Economic Reform in Ukraine: The Unfinished Agenda*. Washington, D.C.: IMF Institute.

KUPETS, O. (2006). Determinants of Unemployment Duration in Ukraine. *Journal of Comparative Economics*.

KUZIO, T. (2013). Yushchenko versus Tymoshenko: Why Ukraine's National Democrats Are Divided. *Demokratyzatsiya*, 21(2), pp. 215–240.

LERMAN, Z. (1999). *Land Reform and Farm Restructuring in Ukraine*. Washington, D.C.: World Bank.

MACROTRENDS. (2024). *Country Trade Balance Data* [online]. Available at: <https://www.macrotrends.net/> [cited 2025-02-09].

MALYNOVSKYY, V. (2024). The Reform of Local Self-Government in Ukraine in the Context of the Polish Experience. *Barometr Regionalny Analizy i Prognozy*.

MINAKOV, M. (2023). *War, De-oligarchization, and the Possibility of Anti-Patronal Transformation in Ukraine*. Washington, D.C.: Kennan Institute.

MINAKOV, M., ROJANSKY, M. (2015). *Democracy in Ukraine: Are We There Yet?* Washington, D.C.: Carnegie Endowment for International Peace.

MYKHENKO, V., SWAIN, A. (2010). Ukraine's Diverging Space-Economy: The Orange Revolution, Post-Soviet Development Models and Regional Trajectories. *European Urban and Regional Studies*.

NOWIAK, W. (2012). Polish "Welfare State" After Transformation: Conclusions and Recommendations for Ukraine.

NORTH, D. C. (1990). *Institutions, Institutional Change and Economic Performance*. Cambridge: Cambridge University Press.

OCIECZEK, G. (2023). Combating Corruption Crime in Poland: Selected Legal Aspects, with Particular Reference to the Role of the Central Anticorruption Bureau. *Law, Identity and Values*.

- PASKHAVER, A., VERKHOVODOVA, L. (2007). Privatization Before and After the Orange Revolution. *Problems of Economic Transition*.
- PYATNYTSKA, G. T., KUCHERENKO, D. O., PAVLENKO, I. S. (2020). Key Success Factors in Domestic Trade: The Experience of Ukraine and Poland.
- SAVASTIEIEVA, O., BUTENKO, V., ZHURAVLOVA, T., FIC, D. (2019). Analysis of Basic Conditions for Implementation of Decentralization Reform in Ukraine and Poland. *Financial and Credit Activity: Problems of Theory and Practice*.
- SERGI, B. S. (2020). *Economic Dynamics in Transitional Economies: The Four-P Governments, the EU Enlargement, and the Bruxelles Consensus*. London: Routledge.
- SLIUSARCHUK, O., LAVROV, R., KUYBIDA, V., SLATVINSKYI, M., ZELENSKYI, A. (2021). Fiscal Policy of Economic Development: Comparative Characteristics of Ukraine and Poland. *Journal of Economic Studies*.
- STAŃCZYK, Z. (2020). *The Success of the Polish Economic Transition*. Warsaw: Polish Economic Journal.
- SURDEJ, A. (2000). *Small- and Medium-Sized Enterprise Development in Poland after 1990*. Warsaw: Warsaw School of Economics.
- SVEJNAR, J. (2002). Transition Economies: Performance and Challenges. *Journal of Economic Perspectives*.
- TRANSPARENCY INTERNATIONAL. (2024). *Corruption Perceptions Index* [online]. Available at: <https://www.transparency.org/en/cpi> [cited 2025-02-20].
- THE OBSERVATORY OF ECONOMIC COMPLEXITY. (n.d.). *Poland*. [online]. Available at: <https://oec.world/en/profile/country/pol> [cited 2025-02-20].
- THE OBSERVATORY OF ECONOMIC COMPLEXITY. (n.d.). *Ukraine*. [online]. Available at: <https://oec.world/en/profile/country/ukr> [cited 2025-02-20].
- WISŁA, R., NOWOSAD, A. (2020). *Economic Transformation in Poland and Ukraine*. London: Routledge.
- WOJCIECHOWSKI, L. (2014). Meeting the Maastricht Criteria by Poland: Reflections on Real Convergence and Monetary Union Membership. *Zeszyty Naukowe Uniwersytetu Ekonomicznego w Krakowie*.
- WORLD BANK. (2024). *World Development Indicators Database* [online]. Washington, D.C.: World Bank. Available at: <https://databank.worldbank.org/source/world-development-indicators> [cited 2025-03-01].
- WORLD BANK. (2024). *Worldwide Governance Indicators (WGI)* [online]. Washington, D.C.: World Bank. Available at: <https://info.worldbank.org/governance/wgi/> [cited 2025-03-01].
- WORLD BANK GROUP. (2020). *Doing Business 2020: Comparing Business Regulation in 190 Economies* [online]. Washington, D.C.: World Bank. Available at: <https://www.doingbusiness.org/> [cited 2025-03-01].
- WOŹNIAK, M., FIRSZT, D. (2014). Development Integration via Real and Technological Convergence: Experience of Poland and Conclusions for Ukraine.
- YEVHEN, I. (2023). Ukraine's Foreign Trade in a Year of War: A Closer Look at the Results of 2022.
- ŽÍDEK, L. (2011). Transformation in Poland: Evaluation and Lessons. *Czech Journal of Economics and Finance*.
- ZIMNY, Z. (2012). Inward FDI in Poland and Its Policy Context. *PSN: Foreign Direct Investment (Development)*.

List of figures

Fig. 1 Development of GDP per capita in Ukraine and Poland during the period 1990 – 2023, expressed in constant 2015 prices

Fig. 2 GDP per capita of Ukraine for the period 1990 – 2023, expressed in constant 2015 prices

Fig. 3. GDP per capita of Poland for the period 1990 – 2023, expressed in constant 2015 prices

Fig. 4 Inflation in Ukraine and Poland in % annual change, for the period 1993-2023

Fig. 5 Development of inflation rate in Ukraine for the period 1993-1996

Fig. 6 Development of inflation rate in Ukraine for the period 1997-2023

Fig. 7 Development of inflation rate in Poland for the period 1993-2023

Fig. 8 Unemployment rates in Ukraine and Poland during the period 1991 – 2021, expressed as the % of total labour force

Fig. 9 Development of unemployment rate in Ukraine during the period 1991 – 2021, as a % of total labor force

Fig. 10 Development of unemployment rate in Poland during the period 1991 – 2021, as a % of total labor force

Fig. 11 Corruption perception index development in Ukraine and Poland during the period 2012-2024

Fig. 12 Corruption perception index development in Ukraine during the period 2012-2024

Fig. 13 Corruption perception index development in Poland during the period 2012-2024

Fig. 14 Control of Corruption estimate index development in Ukraine and Poland during the period 1996-2023

Fig. 15 Control of Corruption estimate index development in Ukraine during the period 1996-2023

Fig. 16 Control of Corruption estimate index development in Poland during the period 1996-2023

Fig.17 Dynamics of the Control of Corruption Rank in Ukraine and Poland 1996–2022

Fig. 18 Government Effectiveness estimate index development in Ukraine and Poland during the period 1996-2023

Fig. 19 Government Effectiveness estimate index development in Ukraine during the period 1996-2023

Fig. 20 Government Effectiveness estimate index development in Poland during the period 1996-2023

Fig.21 Dynamics of the Government Effectiveness Rank in Ukraine and Poland 1996–2022

Fig. 22 Regulatory Quality estimate index development in Ukraine and Poland during the period 1996-2023

Fig. 23 Regulatory Quality estimate index development in Ukraine during the period 1996-2023

Fig. 24 Regulatory Quality estimate index development in Poland during the period 1996-2023

Fig.25 Dynamics of the Regulatory Quality Rank in Ukraine and Poland 1996–2022

Fig.26 Dynamics of the Voice and Accountability Rank in Ukraine and Poland 1996–2022

Fig.27 Dynamics of the Political Stability No Violence Rank in Ukraine and Poland 1996–2022

Fig.28 Dynamics of the Rule of Law Rank in Ukraine and Poland 1996–2022

Fig. 29 Development of Trade Balance in Ukraine and Poland (as a % of the GDP) during the period 1995 – 2023

Fig. 30 Development of Trade Balance in Ukraine (as a % of the GDP) during the period 1995 – 2023

Fig. 31 Development of Trade Balance in Poland (as a % of the GDP) during the period 1995 –2023

Fig. 32 Development of FDI inflows in Ukraine and Poland (as a % of the GDP) during the period 1992 – 2023

Fig. 33 Development of FDI inflows in Ukraine (as a % of the GDP) during the period 1992 – 2023

Fig. 34 Development of FDI inflows in Poland (as a % of the GDP) during the period 1992 – 2023

Fig. 35 Ukraine's FDI inflows by sectors for 2010 and 2018

Fig. 36 Poland's FDI inflows by sectors for 2010 and 2018

Fig. 37 Index chart for GDP per capita dynamics comparison for Ukraine and Poland (1995=100)

Fig. 38 Index chart for FDI inflows dynamics comparison for Ukraine and Poland (1995=100)

Fig. 39 Index chart for Government Effectiveness indicator dynamics comparison for Ukraine and Poland (2002=100)

Fig. 40 Index chart for Control of Corruption indicator dynamics comparison for Ukraine and Poland (2002=100)

Fig. 41 Scatter plot for Correlation between Economic Growth and Government Effectiveness for Ukraine and Poland (2002–2023)

Fig. 42 Scatter plot for Correlation between Economic Growth and Control of Corruption for Ukraine and Poland (2002–2023)

Fig. 43 Econometric model for time series expressing GDP per capita in Ukraine

Fig. 44 Econometric model for time series expressing GDP per capita in Poland

Fig. 45 Econometric model for time series expressing inflation in Ukraine (as a % annual change)

Fig. 46 Econometric model for time series expressing inflation in Poland (as a % annual change)

Fig. 47 Econometric model for time series expressing unemployment in Ukraine (as a % of total labor force)

Fig. 48 Econometric model for time series expressing unemployment in Poland (as a % of total labor force)

Fig. 49 Econometric model for time series of Corruption perception index in Ukraine

Fig. 50 Econometric model for time series of Corruption perception index in Poland

Fig. 51 Econometric model for time series of Control of Corruption estimate index in Ukraine

Fig. 52 Econometric model for time series of Control of Corruption estimate index in Poland

Fig. 53 Econometric model for time series of Governmate Effectiveness estimate index in Ukraine

Fig. 54 Econometric model for time series of Governmate Effectiveness estimate index in Poland

Fig. 55 Econometric model for time series of Regulatory Quality estimate index in Ukraine

Fig. 56 Econometric model for time series of Regulatory Quality estimate index in Poland

Fig. 57 Econometric model for time series expressing trade balance in Ukraine (as a % of GDP)

Fig. 58 Econometric model for time series expressing trade balance in Poland (as a % of GDP)

Fig. 59 Econometric model for time series expressing FDI inflows in Ukraine (as a % of GDP)

Fig. 60 Econometric model for time series expressing FDI inflows in Poland (as a % of GDP)

List of tables

Tab. 1 Summary of the reviewed literature for the part 1.1

Tab. 2 Summary of the reviewed literature for the part 1.2

Tab. 3 Summary of the reviewed literature for the part 1.3

Tab. 4 Summary of the reviewed literature for the part 1.4

Tab. 5 Summary of data used in the research

Tab. 6 Summary regarding the models of time series for the basic economic indicators

Tab. 7 Summary regarding the models of time series for institutional quality and governance indicators

Tab. 8 Comparison of Key Export Goods and Main Trade Partners: Ukraine and Poland (1995; 2010; 2023)

Tab. 9 Summary regarding the models of time series for trade and investment patterns indicators

Tab. 10 Summary comparison table for all analyzed indicators

Tab. 11 Data used in part 3.1

Tab. 12 Corruption Perception Index data

Tab. 13 Worldwide Governance Indicators estimate index data

Tab. 14 Worldwide Governance Indicators rank data

Tab. 15 Data used in part 3.3

Tab. 16 Ukraine's FDI inflows by sectors: 2010 vs. 2018

Tab. 17 Poland's FDI inflows by sectors: 2010 vs. 2018

Attachments

A Tables with data used in the empirical part

Tab. 11 Data used in part 3.1

GDP per capita, expressed in constant 2015 \$ prices					
Year	Ukraine	Poland	Year	Ukraine	Poland
1990	3102.3	5111.3	2007	2498.4	9760.7
1991	2826.1	4735.9	2008	2567.7	10187.2
1992	2538.9	4840.2	2009	2188.2	10446.5
1993	2177.2	5008.3	2010	2285.9	10808.4
1994	1687.5	5262.3	2011	2418.1	11370.3
1995	1493.4	5671.9	2012	2426.9	11542.2
1996	1355.6	6013.0	2013	2432.4	11628.2
1997	1326.3	6391.1	2014	2308.6	12093.2
1998	1311.6	6684.1	2015	2094.5	12637.5
1999	1319.5	6998.8	2016	2156.1	13026.2
2000	1409.2	7401.6	2017	2217.8	13695.6
2001	1547.3	7495.0	2018	2308.7	14551.1
2002	1644.3	7641.0	2019	2398.1	15221.3
2003	1814.3	7915.6	2020	2324.1	14937.6
2004	2042.4	8323.4	2021	2426.6	16368.6
2005	2122.6	8598.6	2022	1874.7	17303.6
2006	2298.1	9137.7	2023	2159.9	17391.1
Inflation, % annual change in consumer prices					
Year	Ukraine	Poland	Year	Ukraine	Poland
1993	4734.9	37.0	2009	15.9	3.8
1994	891.2	33.0	2010	9.4	2.6
1995	376.7	28.0	2011	8.0	4.2
1996	80.3	19.8	2012	0.6	3.6
1997	15.9	14.9	2013	-0.2	1.0
1998	10.6	11.6	2014	12.1	0.1
1999	22.7	7.2	2015	48.7	-0.9
2000	28.2	9.9	2016	13.9	-0.7
2001	12.0	5.4	2017	14.4	2.1
2002	0.8	1.9	2018	11.0	1.8
2003	5.2	0.7	2019	7.9	2.2
2004	9.0	3.4	2020	2.7	3.4
2005	13.6	2.2	2021	9.4	5.1
2006	9.1	1.3	2022	20.2	14.4
2007	12.8	2.5	2023	12.8	11.5
2008	25.2	4.2			
Unemployment rate, as a % of total labor force					
Year	Ukraine	Poland	Year	Ukraine	Poland
1991	1.9	13.6	2007	6.4	9.6

1992	1.9	13.6	2008	6.4	7.1
1993	2.0	14.4	2009	8.8	8.2
1994	2.0	14.8	2010	8.1	9.6
1995	5.6	13.7	2011	7.9	9.6
1996	7.7	12.7	2012	7.5	10.1
1997	8.9	11.0	2013	7.2	10.3
1998	11.3	9.9	2014	9.3	9.0
1999	11.9	12.3	2015	9.1	7.5
2000	11.7	16.3	2016	9.4	6.2
2001	11.1	18.4	2017	9.5	4.9
2002	10.1	19.9	2018	8.8	3.8
2003	9.1	19.4	2019	8.2	3.3
2004	8.6	19.1	2020	9.5	3.2
2005	7.2	17.7	2021	9.8	3.4
2006	6.8	13.8			

Data Source: World Bank

Tab. 12 Corruption Perception Index data

Corruption perception index					
Year	Ukraine	Poland	Year	Ukraine	Poland
2012	26	58	2019	30	58
2013	25	60	2020	33	56
2014	26	61	2021	32	56
2015	27	63	2022	33	55
2016	29	62	2023	36	54
2017	30	60	2024	35	53
2018	32	60			

Data Source: Transparency International

Tab. 13 Worldwide Governance Indicators estimate index data

Control of Corruption estimate index					
Year	Ukraine	Poland	Year	Ukraine	Poland
1996	-1.11	0.71	2010	-1.03	0.51
1997	-1.18	0.75	2011	-1.09	0.56
1998	-1.26	0.79	2012	-1.12	0.66
1999	-1.18	0.74	2013	-1.18	0.65
2000	-1.11	0.70	2014	-1.02	0.68
2001	-1.10	0.54	2015	-1.04	0.69
2002	-1.09	0.38	2016	-0.87	0.76
2003	-0.97	0.40	2017	-0.83	0.70
2004	-0.98	0.11	2018	-0.92	0.62
2005	-0.74	0.25	2019	-0.80	0.61
2006	-0.77	0.26	2020	-0.83	0.62

2007	-0.82	0.28	2021	-0.79	0.55
2008	-0.86	0.45	2022	-0.63	0.51
2009	-1.07	0.44	2023	-0.69	0.57
Government Effectiveness estimate index					
Year	Ukraine	Poland	Year	Ukraine	Poland
1996	-0.67	0.68	2010	-0.82	0.63
1997	-0.76	0.65	2011	-0.87	0.60
1998	-0.85	0.62	2012	-0.59	0.66
1999	-0.76	0.60	2013	-0.66	0.66
2000	-0.68	0.57	2014	-0.40	0.77
2001	-0.69	0.51	2015	-0.55	0.74
2002	-0.70	0.45	2016	-0.61	0.65
2003	-0.67	0.51	2017	-0.51	0.55
2004	-0.67	0.44	2018	-0.46	0.55
2005	-0.67	0.45	2019	-0.33	0.51
2006	-0.52	0.33	2020	-0.41	0.32
2007	-0.72	0.36	2021	-0.44	0.25
2008	-0.79	0.46	2022	-0.50	0.26
2009	-0.84	0.52	2023	-0.36	0.42
Regulatory Quality estimate index					
Year	Ukraine	Poland	Year	Ukraine	Poland
1996	-0.42	0.72	2010	-0.45	1.02
1997	-0.56	0.67	2011	-0.56	0.96
1998	-0.71	0.62	2012	-0.55	1.00
1999	-0.56	0.66	2013	-0.59	1.05
2000	-0.41	0.70	2014	-0.59	1.06
2001	-0.51	0.68	2015	-0.62	0.95
2002	-0.62	0.65	2016	-0.41	0.91
2003	-0.56	0.69	2017	-0.28	0.80
2004	-0.35	0.76	2018	-0.26	0.87
2005	-0.51	0.81	2019	-0.22	1.01
2006	-0.49	0.70	2020	-0.27	0.85
2007	-0.41	0.77	2021	-0.29	0.83
2008	-0.52	0.84	2022	-0.33	0.72
2009	-0.56	0.98	2023	-0.27	0.78

Data Source: World Bank

Tab. 14 Worldwide Governance Indicators rank data

Control of Corruption Rank					
Year	Ukraine	Poland	Year	Ukraine	Poland
1996	13.44	75.81	2010	16.19	71.90
1998	4.27	77.54	2012	12.32	72.99
2000	11.17	73.4	2014	14.42	72.60
2002	9.52	68.78	2016	19.05	75.71
2004	16.26	58.13	2018	17.62	73.33

2006	24.88	63.41	2020	22.86	72.38
2008	19.9	70.87	2022	29.25	68.40
Government Effectiveness Rank					
Year	Ukraine	Poland	Year	Ukraine	Poland
1996	27.87	75.41	2010	23.92	70.81
1998	22.40	72.13	2012	32.70	70.62
2000	27.32	69.40	2014	39.90	74.52
2002	29.19	66.49	2016	31.43	72.38
2004	30.35	67.16	2018	37.43	71.90
2006	36.59	64.39	2020	37.62	64.29
2008	22.82	67.48	2022	33.02	61.79
Regulatory Quality Rank					
Year	Ukraine	Poland	Year	Ukraine	Poland
1996	33.70	72.28	2010	35.71	81.34
1998	23.37	70.65	2012	31.28	79.15
2000	34.24	72.83	2014	31.25	81.25
2002	29.19	69.19	2016	38.57	79.52
2004	42.29	73.13	2018	43.33	78.57
2006	33.82	71.57	2020	42.38	76.19
2008	33.98	74.76	2022	40.57	74.53
Voice and Accountability Rank					
Year	Ukraine	Poland	Year	Ukraine	Poland
1996	39.50	80.00	2010	45.02	80.09
1998	43.28	81.09	2012	40.38	82.63
2000	30.85	81.09	2014	43.35	82.27
2002	33.33	80.60	2016	47.78	74.38
2004	28.37	82.21	2018	43.69	70.87
2006	47.60	73.56	2020	51.69	66.18
2008	50.48	75.96	2022	45.89	65.22
Political Stability No Violence Rank					
Year	Ukraine	Poland	Year	Ukraine	Poland
1996	42.55	73.94	2010	45.50	84.36
1998	39.89	74.47	2012	41.71	86.26
2000	32.28	59.26	2014	5.71	76.19
2002	34.39	71.43	2016	6.67	63.33
2004	29.61	50.97	2018	5.66	62.74
2006	43.48	55.07	2020	12.26	62.26
2008	45.67	77.88	2022	5.66	61.79
Rule of Law Rank					
Year	Ukraine	Poland	Year	Ukraine	Poland
1996	22.61	70.85	2010	23.70	68.72
1998	18.50	72.50	2012	24.41	73.24
2000	15.92	69.15	2014	20.67	78.37
2002	23.88	67.66	2016	20.00	71.43
2004	24.52	61.54	2018	23.81	63.81
2006	24.88	62.20	2020	25.24	67.62

2008	27.40	66.83	2022	18.87	64.15
-------------	-------	-------	-------------	-------	-------

Data Source: World Bank

Tab. 15 Data used in part 3.3

Trade Balance, as a % of GDP					
Year	Ukraine	Poland	Year	Ukraine	Poland
1995	-3.08	2.16	2010	-2.82	-2.37
1996	-2.55	-1.41	2011	-5.99	-2.14
1997	-3.06	-3.88	2012	-7.86	-0.71
1998	-2.27	-4.84	2013	-8.21	1.18
1999	5.46	-5.93	2014	-3.51	0.41
2000	4.86	-6.49	2015	-2.61	2.08
2001	1.37	-3.78	2016	-6.92	3
2002	4.24	-3.56	2017	-7.79	2.83
2003	2.5	-2.75	2018	-8.73	2.05
2004	7.41	-2.97	2019	-8.05	3.7
2005	0.81	-1.31	2020	-1.51	5.66
2006	-2.74	-2.33	2021	-1.32	3.32
2007	-5.47	-3.8	2022	-16.51	1.78
2008	-7.65	-5.28	2023	-20.93	6.09
2009	-1.62	-0.98			
FDI inflows, as a % of GDP					
Year	Ukraine	Poland	Year	Ukraine	Poland
1992	0.3	0.7	2008	5.7	2.7
1993	0.3	1.8	2009	3.9	3.2
1994	0.3	1.7	2010	4.6	3.9
1995	0.6	2.6	2011	4.3	3.5
1996	1.2	2.8	2012	4.5	1.4
1997	1.2	3.1	2013	2.4	0.2
1998	1.8	3.7	2014	0.6	3.8
1999	1.6	4.4	2015	-0.2	3.3
2000	1.8	5.4	2016	4.4	3.8
2001	2.0	3.0	2017	3.3	2.3
2002	1.6	2.1	2018	3.8	3.3
2003	2.7	2.5	2019	3.8	3.0
2004	2.6	5.4	2020	0.2	3.2
2005	8.7	3.6	2021	4.0	5.3

2006	5.0	6.2	2022	0.1	5.3
2007	6.9	5.8	2023	2.7	3.9

Data Source: World Bank and Macrotrends

Tab. 16 Ukraine`s FDI inflows by sectors: 2010 vs. 2018

Sector	Ukraine 2010	Ukraine 2018
Manufacturing	14,5	12,9
Financial sector	32,1	24,3
ICT and Telecom	3,2	8,2
Real estate	10,4	13,1
Trade	17,8	16,9
Energy	9,3	6,3
Transport and logistics	2,6	4
Agro-industrial sector	2,1	5,1
Other	8	9,2

Data source: German Economic Team

Tab. 17 Poland`s FDI inflows by sectors: 2010 vs. 2018

Sector	Poland 2010	Poland 2018
Manufacturing	21,7	28,4
Financial sector	14,8	10,2
ICT and Telecom	4	6,7
Real estate	10,6	12,5
Trade	9,4	7,1
Energy	5,2	3,3
Transport and logistics	8,9	11,6
Agro-industrial sector	4,3	2,5
Other	21,1	17,7

Data source: German Economic Team

B Econometric models used in empirical part

Fig. 43 Econometric model for time series expressing GDP per capita in Ukraine

Model 2: OLS, using observations 1991-2023 (T = 33)					
Dependent variable: Ukraine					
	<i>Coefficient</i>	<i>Std. Error</i>	<i>t-ratio</i>	<i>p-value</i>	
const	2869.68	186.709	15.37	<0.0001	***
time	-177.209	28.7810	-6.157	<0.0001	***
break2000	-1214.40	240.549	-5.048	<0.0001	***
break2000t	199.182	29.4775	6.757	<0.0001	***
uhat1_1	0.599106	0.155757	3.846	0.0006	***
Mean dependent var	2060.632	S.D. dependent var		427.7096	
Sum squared resid	1352809	S.E. of regression		219.8058	
R-squared	0.768906	Adjusted R-squared		0.735893	
F(4, 28)	23.29073	P-value(F)		1.46e-08	
Log-likelihood	-222.0745	Akaike criterion		454.1491	
Schwarz criterion	461.6316	Hannan-Quinn		456.6667	
rho	0.078304	Durbin-Watson		1.817589	

Fig. 44 Econometric model for time series expressing GDP per capita in Poland

Model 4: OLS, using observations 1991-2023 (T = 33)					
Dependent variable: Poland					
	<i>Coefficient</i>	<i>Std. Error</i>	<i>t-ratio</i>	<i>p-value</i>	
const	3667.13	108.053	33.94	<0.0001	***
time	332.903	7.40399	44.96	<0.0001	***
break2014	-7051.73	776.205	-9.085	<0.0001	***
break2014t	280.711	26.9718	10.41	<0.0001	***
uhat3_1	0.358967	0.136927	2.622	0.0140	**
Mean dependent var	10032.97	S.D. dependent var		3794.171	
Sum squared resid	1553117	S.E. of regression		235.5174	
R-squared	0.996629	Adjusted R-squared		0.996147	
F(4, 28)	2069.240	P-value(F)		3.67e-34	
Log-likelihood	-224.3529	Akaike criterion		458.7057	
Schwarz criterion	466.1883	Hannan-Quinn		461.2234	
rho	-0.024705	Durbin-Watson		2.007207	

Fig. 45 Econometric model for time series expressing inflation in Ukraine (as a % annual change)

Model 4: OLS, using observations 1994-2023 (T = 30)					
Dependent variable: Ukraine					
	<i>Coefficient</i>	<i>Std. Error</i>	<i>t-ratio</i>	<i>p-value</i>	
const	1455.37	82.6671	17.61	<0.0001	***
time	-315.287	34.8780	-9.040	<0.0001	***
break1996	-1441.33	81.2087	-17.75	<0.0001	***

break1996t	315.232	34.8063	9.057	<0.0001	***
uhat3_1	0.119676	0.0191556	6.248	<0.0001	***
Mean dependent var	56.67000	S.D. dependent var	171.5461		
Sum squared resid	2496.627	S.E. of regression	9.993251		
R-squared	0.997075	Adjusted R-squared	0.996606		
F(4, 25)	2130.168	P-value(F)	2.86e-31		
Log-likelihood	-108.8906	Akaike criterion	227.7813		
Schwarz criterion	234.7872	Hannan-Quinn	230.0225		
rho	0.150947	Durbin-Watson	1.697859		

Fig. 46 Econometric model for time series expressing inflation in Poland (as a % annual change)

Model 2: OLS, using observations 1994-2023 (T = 30)
Dependent variable: Poland

	<i>Coefficient</i>	<i>Std. Error</i>	<i>t-ratio</i>	<i>p-value</i>	
const	36.4127	2.32124	15.69	<0.0001	***
time	-3.61767	0.391529	-9.240	<0.0001	***
break2002	-39.7732	2.93580	-13.55	<0.0001	***
break2002t	3.94171	0.398317	9.896	<0.0001	***
uhat1_1	0.718758	0.163734	4.390	0.0002	***
Mean dependent var	6.673333	S.D. dependent var	8.209495		
Sum squared resid	157.5813	S.E. of regression	2.510628		
R-squared	0.919374	Adjusted R-squared	0.906474		
F(4, 25)	71.26867	P-value(F)	2.68e-13		
Log-likelihood	-67.44932	Akaike criterion	144.8986		
Schwarz criterion	151.9046	Hannan-Quinn	147.1399		
rho	-0.030396	Durbin-Watson	2.010095		

Fig. 47 Econometric model for time series expressing unemployment in Ukraine (as a % of totallabor force)

Model 2: OLS, using observations 1991-2021 (T = 31)
Dependent variable: Ukraine

	<i>Coefficient</i>	<i>Std. Error</i>	<i>t-ratio</i>	<i>p-value</i>	
const	-0.552727	0.798809	-0.6919	0.4949	
time	1.24364	0.117778	10.56	<0.0001	***
break2002	7.46829	1.33231	5.606	<0.0001	***
break2002t	-1.17529	0.127146	-9.244	<0.0001	***
Mean dependent var	7.861290	S.D. dependent var	2.768234		
Sum squared resid	41.19882	S.E. of regression	1.235266		
R-squared	0.820792	Adjusted R-squared	0.800880		
F(3, 27)	41.22091	P-value(F)	3.24e-10		
Log-likelihood	-48.39564	Akaike criterion	104.7913		
Schwarz criterion	110.5272	Hannan-Quinn	106.6611		
rho	0.317149	Durbin-Watson	1.325013		

Fig. 48 Econometric model for time series expressing unemployment in Poland (as a % of totallabor force)

Model 7: OLS, using observations 1992-2021 (T = 30)					
Dependent variable: Poland					
	<i>Coefficient</i>	<i>Std. Error</i>	<i>t-ratio</i>	<i>p-value</i>	
const	17.1682	0.657093	26.13	<0.0001	***
time	-0.758073	0.0549751	-13.79	<0.0001	***
break2000	8.80483	1.07914	8.159	<0.0001	***
uhat1_1	0.627568	0.150887	4.159	0.0003	***
Mean dependent var	11.09333	S.D. dependent var		5.027506	
Sum squared resid	71.22390	S.E. of regression		1.655107	
R-squared	0.902832	Adjusted R-squared		0.891620	
F(3, 26)	80.52608	P-value(F)		2.75e-13	
Log-likelihood	-55.53762	Akaike criterion		119.0752	
Schwarz criterion	124.6800	Hannan-Quinn		120.8683	
rho	0.212777	Durbin-Watson		1.564744	

Fig. 49 Econometric model for time series of Corruption perception index in Ukraine

Model 1: OLS, using observations 2012-2024 (T = 13)					
Dependent variable: Ukraine					
	<i>Coefficient</i>	<i>Std. Error</i>	<i>t-ratio</i>	<i>p-value</i>	
const	24.1538	0.617317	39.13	<0.0001	***
time	0.879121	0.0777746	11.30	<0.0001	***
Mean dependent var	30.30769	S.D. dependent var		3.568020	
Sum squared resid	12.10989	S.E. of regression		1.049237	
R-squared	0.920731	Adjusted R-squared		0.913525	
F(1, 11)	127.7677	P-value(F)		2.15e-07	
Log-likelihood	-17.98518	Akaike criterion		39.97035	
Schwarz criterion	41.10025	Hannan-Quinn		39.73811	
rho	-0.318786	Durbin-Watson		2.514479	

Fig. 50 Econometric model for time series of Corruption perception index in Poland

Model 3: OLS, using observations 2012-2024 (T = 13)					
Dependent variable: Poland					
	<i>Coefficient</i>	<i>Std. Error</i>	<i>t-ratio</i>	<i>p-value</i>	
const	56.5000	0.644061	87.72	<0.0001	***
time	1.60000	0.235178	6.803	<0.0001	***
break2016	10.5111	0.904917	11.62	<0.0001	***
break2016t	-2.70000	0.244781	-11.03	<0.0001	***
Mean dependent var	58.15385	S.D. dependent var		3.158221	
Sum squared resid	2.488889	S.E. of regression		0.525874	
R-squared	0.979206	Adjusted R-squared		0.972275	

F(3, 9)	141.2720	P-value(F)	6.91e-08
Log-likelihood	-7.700967	Akaike criterion	23.40193
Schwarz criterion	25.66173	Hannan-Quinn	22.93744
rho	-0.171885	Durbin-Watson	2.294692

Fig. 51 Econometric model for time series of Control of Corruption estimate index in Ukraine

Model 3: OLS, using observations 1996-2023 (T = 28)
Dependent variable: Ukraine

	<i>Coefficient</i>	<i>Std. Error</i>	<i>t-ratio</i>	<i>p-value</i>	
const	-1.26023	0.0336509	-37.45	<0.0001	***
time	0.0351753	0.00365990	9.611	<0.0001	***
break2009	-0.405461	0.0592789	-6.840	<0.0001	***
Mean dependent var	-0.967400	S.D. dependent var		0.169864	
Sum squared resid	0.154711	S.E. of regression		0.078667	
R-squared	0.801411	Adjusted R-squared		0.785524	
F(2, 25)	50.44413	P-value(F)		1.68e-09	
Log-likelihood	33.04738	Akaike criterion		-60.09475	
Schwarz criterion	-56.09814	Hannan-Quinn		-58.87295	
rho	0.218985	Durbin-Watson		1.476564	

Fig. 52 Econometric model for time series of Control of Corruption estimate index in Poland

Model 8: OLS, using observations 1997-2023 (T = 27)
Dependent variable: Poland

	<i>Coefficient</i>	<i>Std. Error</i>	<i>t-ratio</i>	<i>p-value</i>	
const	0.916647	0.0660007	13.89	<0.0001	***
time	-0.0628361	0.00863831	-7.274	<0.0001	***
break2008	-0.427645	0.121468	-3.521	0.0019	***
break2008t	0.0679130	0.00989207	6.865	<0.0001	***
uhat7_1	0.461690	0.186727	2.473	0.0216	**
Mean dependent var	0.546863	S.D. dependent var		0.177123	
Sum squared resid	0.176897	S.E. of regression		0.089670	
R-squared	0.783131	Adjusted R-squared		0.743701	
F(4, 22)	19.86097	P-value(F)		4.80e-07	
Log-likelihood	29.56696	Akaike criterion		-49.13392	
Schwarz criterion	-42.65474	Hannan-Quinn		-47.20732	
rho	0.070587	Durbin-Watson		1.857927	

Fig. 53 Econometric model for time series of Governmate Effectiveness estimate index in Ukraine

Model 3: OLS, using observations 1996-2023 (T = 28)
Dependent variable: Ukraine

<i>Coefficient</i>	<i>Std. Error</i>	<i>t-ratio</i>	<i>p-value</i>
--------------------	-------------------	----------------	----------------

const	-0.854299	0.0352450	-24.24	<0.0001	***
time	0.0267498	0.00386653	6.918	<0.0001	***
break2007	-0.261501	0.0639511	-4.089	0.0004	***
Mean dependent var	-0.625196	S.D. dependent var		0.155102	
Sum squared resid	0.193603	S.E. of regression		0.088001	
R-squared	0.701935	Adjusted R-squared		0.678089	
F(2, 25)	29.43710	P-value(F)		2.68e-07	
Log-likelihood	29.90784	Akaike criterion		-53.81568	
Schwarz criterion	-49.81907	Hannan-Quinn		-52.59388	
rho	0.190551	Durbin-Watson		1.490671	

Fig. 54 Econometric model for time series of Governmate Effectiveness estimate index in Poland

Model 9: OLS, using observations 1997-2023 (T = 27)
Dependent variable: Poland

	<i>Coefficient</i>	<i>Std. Error</i>	<i>t-ratio</i>	<i>p-value</i>	
const	0.684298	0.0399019	17.15	<0.0001	***
time	-0.0223789	0.00420534	-5.322	<0.0001	***
break2010	0.333130	0.0660767	5.042	<0.0001	***
uhat6_1	0.537102	0.191606	2.803	0.0101	**
Mean dependent var	0.520402	S.D. dependent var		0.137448	
Sum squared resid	0.151333	S.E. of regression		0.081115	
R-squared	0.691908	Adjusted R-squared		0.651722	
F(3, 23)	17.21766	P-value(F)		4.41e-06	
Log-likelihood	31.67417	Akaike criterion		-55.34834	
Schwarz criterion	-50.16499	Hannan-Quinn		-53.80705	
rho	0.244954	Durbin-Watson		1.472747	

Fig. 55 Econometric model for time series of Regulatory Quality estimate index in Ukraine

Model 5: OLS, using observations 1997-2023 (T = 27)
Dependent variable: Ukraine

	<i>Coefficient</i>	<i>Std. Error</i>	<i>t-ratio</i>	<i>p-value</i>	
const	-0.630600	0.0390396	-16.15	<0.0001	***
time	0.0113819	0.00231073	4.926	<0.0001	***
uhat1_1	0.480807	0.167443	2.871	0.0084	***
Mean dependent var	-0.460918	S.D. dependent var		0.136570	
Sum squared resid	0.209627	S.E. of regression		0.093458	
R-squared	0.567720	Adjusted R-squared		0.531697	
F(2, 24)	15.75980	P-value(F)		0.000043	
Log-likelihood	27.27523	Akaike criterion		-48.55046	
Schwarz criterion	-44.66295	Hannan-Quinn		-47.39450	
rho	0.143748	Durbin-Watson		1.700706	

Fig. 56 Econometric model for time series of Regulatory Quality estimate index in Poland

Model 7: OLS, using observations 1996-2023 (T = 28)

Dependent variable: Poland					
	<i>Coefficient</i>	<i>Std. Error</i>	<i>t-ratio</i>	<i>p-value</i>	
const	0.632204	0.0366775	17.24	<0.0001	***
time	0.0117083	0.00462093	2.534	0.0182	**
break2009	0.668563	0.0878929	7.607	<0.0001	***
break2009t	-0.0298271	0.00593569	-5.025	<0.0001	***
Mean dependent var	0.824578	S.D. dependent var		0.136896	
Sum squared resid	0.093270	S.E. of regression		0.062340	
R-squared	0.815669	Adjusted R-squared		0.792628	
F(3, 24)	35.40028	P-value(F)		5.65e-09	
Log-likelihood	40.13219	Akaike criterion		-72.26438	
Schwarz criterion	-66.93557	Hannan-Quinn		-70.63531	
rho	0.108900	Durbin-Watson		1.710550	

Fig. 57 Econometric model for time series expressing trade balance in Ukraine (as a % of GDP)

Model 15: OLS, using observations 1996-2023 (T = 28) Dependent variable: Ukraine					
	<i>Coefficient</i>	<i>Std. Error</i>	<i>t-ratio</i>	<i>p-value</i>	
const	-3.22904	1.67498	-1.928	0.0663	*
time	0.816400	0.275254	2.966	0.0069	***
break2005t	-0.824132	0.219419	-3.756	0.0010	***
break2022	-12.3351	2.20359	-5.598	<0.0001	***
Ukraine_1	0.329279	0.129572	2.541	0.0182	**
Mean dependent var	-3.623929	S.D. dependent var		6.216394	
Sum squared resid	158.4647	S.E. of regression		2.624837	
R-squared	0.848123	Adjusted R-squared		0.821710	
F(4, 23)	32.10961	P-value(F)		4.16e-09	
Log-likelihood	-63.99686	Akaike criterion		137.9937	
Schwarz criterion	144.6547	Hannan-Quinn		140.0301	
rho	0.076426	Durbin's h		0.555548	

Fig. 58 Econometric model for time series expressing trade balance in Poland (as a % of GDP)

Model 17: OLS, using observations 1995-2023 (T = 29) Dependent variable: Poland					
	<i>Coefficient</i>	<i>Std. Error</i>	<i>t-ratio</i>	<i>p-value</i>	
const	3.87500	1.69853	2.281	0.0313	**
time	-2.34700	0.620214	-3.784	0.0009	***
break1999	-11.8730	1.84106	-6.449	<0.0001	***
break1999t	2.78853	0.621406	4.487	0.0001	***
Mean dependent var	-0.698966	S.D. dependent var		3.468343	
Sum squared resid	48.08318	S.E. of regression		1.386841	

R-squared	0.857245	Adjusted R-squared	0.840114
F(3, 25)	50.04176	P-value(F)	1.04e-10
Log-likelihood	-48.48095	Akaike criterion	104.9619
Schwarz criterion	110.4311	Hannan-Quinn	106.6748
rho	0.166436	Durbin-Watson	1.635960

Fig. 59 Econometric model for time series expressing FDI inflows in Ukraine (as a % of GDP)

Model 3: OLS, using observations 1992-2023 (T = 32)
Dependent variable: Ukraine

	<i>Coefficient</i>	<i>Std. Error</i>	<i>t-ratio</i>	<i>p-value</i>	
const	-0.0307692	0.732041	-0.04203	0.9668	
time	0.202198	0.0922285	2.192	0.0375	**
break2005	15.0419	3.01126	4.995	<0.0001	***
break2005t	-0.752198	0.185224	-4.061	0.0004	***
break2014	-14.3578	4.78305	-3.002	0.0059	***
break2014t	0.608788	0.211108	2.884	0.0078	***
Mean dependent var	2.709375	S.D. dependent var		2.146168	
Sum squared resid	40.25081	S.E. of regression		1.244230	
R-squared	0.718106	Adjusted R-squared		0.663896	
F(5, 26)	13.24667	P-value(F)		1.82e-06	
Log-likelihood	-49.07634	Akaike criterion		110.1527	
Schwarz criterion	118.9471	Hannan-Quinn		113.0678	
rho	-0.236799	Durbin-Watson		2.472185	

Fig. 60 Econometric model for time series expressing FDI inflows in Poland (as a % of GDP)

Model 6: OLS, using observations 1992-2023 (T = 32)
Dependent variable: Poland

	<i>Coefficient</i>	<i>Std. Error</i>	<i>t-ratio</i>	<i>p-value</i>	
const	1.15821	0.553006	2.094	0.0451	**
time	0.263425	0.0581600	4.529	<0.0001	***
break2008t	-0.176661	0.0423624	-4.170	0.0003	***
Mean dependent var	3.340625	S.D. dependent var		1.429608	
Sum squared resid	37.10135	S.E. of regression		1.131087	
R-squared	0.414410	Adjusted R-squared		0.374024	
F(2, 29)	10.26134	P-value(F)		0.000427	
Log-likelihood	-47.77271	Akaike criterion		101.5454	
Schwarz criterion	105.9426	Hannan-Quinn		103.0030	
rho	0.215934	Durbin-Watson		1.554078	